

Adolescent Schooling and Adult Labor Supply: Evidence from COVID-19 School Closures and Reopenings in Kenya

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Abstract

This paper identifies the impact of a shock to adolescent schooling—potentially affecting both childcare burdens and child labor—on adult labor supply and intra-household allocation of productive activities in the context of COVID-19-related school closures in Kenya. We compare changes in outcomes after schools partially reopened in October 2020 for households with children in a grade eligible to return against those with children in adjacent grades. Using nationally-representative bi-monthly panel data, we find that a child returning to school increases adults’ weekly work by 4.3 hours (27%) in the short run, concentrated among the most flexible margins of adjustment and particularly household agriculture. Contrary to evidence from high-income settings, overall effects are not gendered. We find no effects of the partial reopening on respondent childcare hours and heterogeneity in labor supply effects by household characteristics generally does not align with predictions based on a childcare mechanism. Instead, the results indicate that increased adult work hours substitute for reduced child work in household agriculture as a child goes back to school. Impacts on labor supply are driven by less wealthy households with children engaged in household agriculture, while wealthier agricultural households substitute child labor with increased hired labor. Our results show that adolescent schooling has important consequences for household production and labor supply decisions. Poor agricultural households face particularly high opportunity costs for children’s education.

JEL codes: D13, J13, J22, J43, O17

Keywords: childcare, child labor, labor supply, gender, Kenya, COVID-19, school closures

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1 Introduction

Children in many low-and middle-income countries (LMICs) play an important role in household productive activities (Rosenzweig and Evenson 1977; Kielland and Tovo 2006). Around 20 percent of children in Sub-Saharan Africa are estimated to engage in child labor, primarily concentrated in household farm work (ILO 2017a; UIS and UNICEF 2017), and more contribute to household production without being formally considered child laborers (Bourdillon 2006). In addition to household production, children also contribute to care of younger siblings (Jakiela et al. 2020; Levison and Moe 1998; Qureshi 2018). These contributions to household production increase the opportunity cost of children attending school, and several studies have explored how households respond to this trade-off.¹ Children’s roles as recipients of childcare and providers of household labor may also affect adult labor supply decisions. In this paper, we study the effects of changes in childcare burdens and child labor availability on adult labor supply in the context of COVID-19 school closures in Kenya.

In 2020, countries around the world closed schools in response to the COVID-19 pandemic. Studies have shown that these closures led to learning loss and other adverse outcomes for affected children, with little evidence that they significantly reduced COVID-19 transmission and morbidity.² In addition, a large literature shows that school closures affected the labor supply of students’ parents.³ These studies fairly consistently find greater reductions in labor supply for mothers with school-age children and point to increases in childcare burdens as the main mechanism.⁴

Most of these studies consider high-income countries, however, and school closures may have had different effects in Sub-Saharan Africa countries. A greater share of labor is concentrated in informal family farm or non-farm enterprise work (ILO 2017b), with women and mothers in particular more likely than men to be engaged in such own-account or contributing family work (Lo Bue et al. 2021). To the extent these activities are more accommodating of childcare needs than wage employment, school closures may have been less of a childcare shock. Use of formal early childhood care is also low (Samman et al. 2016), meaning more care may already have been concentrated within the household. Importantly, unlike children in LMICs, those in high-income countries rarely contribute to household production activities. The COVID-19 school closures reduced the opportunity cost of having children work on the household farm, and increases in child labor and children’s work at home were reported in many countries (Habib et al. 2024). For example, in Uganda the share of children engaged in child labor increased from 21% in 2019 to 36% in 2020 while schools were closed

1. See e.g., Basu and Van (1998), Beegle, R. Dehejia, and Gatti (2009), Levison and Moe (1998), Edmonds (2006), Rosenzweig and Evenson (1977), and Shah and Steinberg (2021).

2. Engzell, Frey, and Verhagen (2021), Felfe et al. (2023), Hume, Brown, and Mahtani (2023), Jack and Oster (2023), Munro et al. (2023), and Singh, Romero, and Muralidharan (2024).

3. See e.g., Albanesi and Kim (2021), Alon et al. (2022), Amuedo-Dorantes et al. (2023), Collins et al. (2021), Couch, Fairlie, and Xu (2022), Del Boca et al. (2020), Furman, Kearney, and Powell (2021), Giurge, Whillans, and Yemiscigil (2021), Hansen, Sabia, and Schaller (2024), Heggeness (2020), Liu, Wei, and Xu (2021), Prados and Zamarro (2021), and Zamarro and Prados (2021).

4. Unlike school breaks or vacations, pandemic closures were generally of undetermined duration and were concurrent with restrictions on alternative sources of childcare, limiting parents’ ability to cope.

(UBOS 2017).

This paper leverages school closure policies in Kenya as exogenous shocks to provide empirical estimates of the impact of childcare responsibilities and child labor availability on adult labor supply and intra-household allocation of productive activities in a LMIC setting. Kenya closed all schools nationwide after its first COVID-19 cases in March 2020, partially reopened schools for specific grades in October 2020, and fully reopened for all grades in January 2021. We exploit quasi-random variation in when children enrolled in different grades were eligible to return to school to implement a difference-in-differences analysis comparing changes in labor supply after the October partial reopening for adults in households with children in grades 4 or 8—eligible to return (99% did)—against those with children in adjacent grades. Our data consists of a nationally representative bi-monthly panel from the Kenya COVID-19 Rapid Response Phone Survey (RRPS), including information on household composition, adult labor, childcare, and child labor. The timing of surveys is ideally suited for our study, with two rounds conducted when schools were fully closed, one round exactly after the partial reopening, and further rounds after all schools reopened.

First, we document childcare and child labor arrangements in Kenya and how these vary with changes in school closure policies. Among sample households, both female and male survey respondents increased childcare in the last 7 days by 15-20 hours while schools were closed relative to the periods before schools closed and after they fully reopened, from a base of 40 hours for women and 25 hours for men. Adult childcare hours (which may be combined with other activities) vary little as the number of household children increases, suggesting significant economies of scale in childcare. We do not have data on sibling childcare during the closures period, but children provided an average of 15 hours of care to siblings in the survey round after schools reopened in 2021.⁵ Very few households report any childcare provided by non-household members, even after the majority of pandemic restrictions were lifted in 2021. Child labor in household agriculture is reported by 38% of agricultural households during the school closures period. Children are reported to have worked 20 hours in total per week on average while schools were closed in 2020 compared to 15 hours in the same months after schools were fully open in 2021. Adults in these households worked a total of 58 hours per week on average. Thus, children in Kenya contribute substantially to both household agricultural production and to sibling childcare within the household.

Next, we analyze impacts of the partial school reopening on adults' labor supply. Effects are concentrated on the intensive margin: weekly work hours increase by 4.3 (27%) after the partial reopening for adults with a child eligible to return to school, driven by a 33% increase in hours worked in household agriculture. We find no effects on wage employment hours, household enterprise hours, or on the extensive margin of employment in the weeks following the reopening.⁶ The partial

5. These results contrast somewhat with a recent study of family structure and childcare arrangements in Sub-Saharan Africa (Donald, Lowes, and Vaillant 2024), which finds that mothers are responsible for most childcare in the DRC with grandmothers providing some support and infrequent care provision by siblings. Differences may be due to the different settings of the studies, or different definitions of 'childcare' in the two studies, with the RRPS employing a very broad definition of childcare.

6. Analysis of longer-term impacts is complicated by the fact that schools fully reopened in January 2021, meaning our comparison group also becomes 'treated.'

reopening does not affect labor supply in households with both children eligible to return to school *and* children in adjacent grades who do not return, where we would expect changes in the childcare burden and in child labor to be muted.

Average labor impacts are driven by agricultural households and adults that were able to continue working during the school closures period. Household agriculture is likely more flexible in allowing adults to adjust working hours and participation in household agriculture was also less affected by the pandemic. Surprisingly, the average impacts of the reopening are not significantly different by sex, contrasting with evidence on pandemic labor supply changes from high-income contexts (see e.g., Alon et al. 2022; Amuedo-Dorantes et al. 2023; Collins et al. 2021; Hansen, Sabia, and Schaller 2024; Heggeness 2020) and expectations based on women’s role as primary caregivers in most Kenyan households. One reason for the lack of difference is that in our sample, both sexes contribute substantially to childcare and increased childcare hours during school closures. The partial reopening may also have been less of a shock to childcare burdens than the school closure policies studied in other papers because the children returning to school were somewhat older and nearly all had younger siblings still at home.

In line with this reasoning, we find no effects of the partial reopening on respondent childcare hours in the last 7 days overall, though some heterogeneity by household characteristics aligns with predictions based on a childcare mechanism. For instance, respondent childcare hours increase by more when there are young children age 0-4 in the household or when the child returning to school is a girl, suggesting older sisters may have helped with childcare when schools were closed. However, heterogeneity in impacts on adult work hours does not consistently align with predictions based on a childcare mechanism. A notable exception is that women in households with only one child significantly increase hours of work outside the household farm when that child returns to school. Thus, while there is some evidence that is consistent with childcare playing a role, it is clear that the childcare mechanism cannot quantitatively explain the large average increases in adult labor supply.

Instead, the evidence indicates that increases in adult labor supply are driven by a need to make up for reductions in child agricultural labor. The partial reopening coincided with the main harvest season for most of Kenya, and adult work hours increase by 7.8 hours in households where children were engaged in household agricultural labor during school closures, compared to no effect in households not reporting child farm work. Treated households also increase the probability of hiring any outside agricultural labor by 5 percentage points (70%), concentrated among wealthier households with children engaged in agricultural labor during school closures. The results indicate that poorer households substitute reduced child labor with adult labor while wealthier households substitute with hired labor: adults in more wealthy households do not increase work hours while those in less wealthy households increase work hours by 8.8 hours. Patterns of heterogeneity by household characteristics are similar for both women and men, indicating increased work hours for both sexes are driven by the child agricultural labor mechanism.

This paper makes three main contributions. First, we consider how formal childcare for school-

age children (through schooling) may affect adults’ labor supply through changes in childcare burdens, in a context where children are often both recipients and providers of childcare. This paper builds on a growing literature identifying causal impacts of childcare availability on mothers’ labor market outcomes in LMICs, generally finding that access to childcare increases labor supply.⁷ We document household childcare arrangements using a nationally-representative sample of Kenyan households with mobile phones, showing significant engagement of men and older siblings and little childcare provision by non-household members. Though the definition of childcare used in the survey is broad, this challenges notions of childcare as overwhelmingly being provided by women in Sub-Saharan Africa (Donald, Lowes, and Vaillant 2024). Using a natural experiment based on changes in school closure policies in Kenya, we show that although adult childcare hours increased while schools were closed and fell after they fully reopened, a single adolescent returning to school does not significantly affect respondent childcare hours.

Second, we demonstrate that a change in children’s labor availability around harvest time affects parents’ labor supply and hired labor. This result is consistent with evidence on the importance of the timing of school breaks for both child and household outcomes in LMICs.⁸ Much of this literature has focused on impacts of school calendar policy on outcomes for children. We show that adults in Kenya make up for reductions in child labor when a student returns to school by increasing their time allocation to household agriculture. This is true particularly for less wealthy households, while wealthier households instead substitute reduced child labor with hired labor.⁹ The result aligns with many studies considering the opportunity cost of children’s school attendance in LMICS.¹⁰ More generally, the results add to a broad literature on child labor in LMICs,¹¹ further demonstrating how children can play an important role in household agricultural production and labor supply decisions.

Finally, we also contribute to understanding labor impacts of pandemics and pandemic-related policies—school closures in particular. Many studies have analyzed the gendered effects of the COVID-19 pandemic on childcare and employment. These primarily report on high-income settings and fairly consistently find that increased childcare burdens contributed to greater adverse labor

7. See Halim, Perova, and Reynolds (2023) for a review of causally identified studies in LMICs and [J-PAL Policy Insight](#) (2023) for another recent review of 9 RCTs of childcare interventions in LMICs. Recent studies from Sub-Saharan African contexts include Ajayi et al. (2024), Bjorvatn et al. (2022), Clark et al. (2019), and Donald and Vaillant (2023). A much broader literature considers this relationship in high-income contexts; see Morrissey (2017) for a recent review.

8. See e.g., Admassie (2003), Allen (2024), Duryea and Arends-Kuenning (2003), Ito, Shonchay, et al. (2020), Kadzamira and Rose (2003), and Merfeld (2024).

9. Allen (2024) reports that households in Malawi respond to harvest-period overlap with the school calendar by increasing expenditure on hired labor, which is not measured in the RRPS, but does not report on household adult labor.

10. See e.g., Bai and Wang (2020), Bau et al. (2020), Edmonds (2006), Rosenzweig and Evenson (1977), and Shah and Steinberg (2021). Doran (2013) finds that Mexico’s PROGRESA program reduced child farm labor and led employers to substitute this with increased hired adult labor, though—contrary to our setting, where household income does not increase immediately from children’s return to school—this may reflect an income effect in addition to a child labor availability mechanism.

11. See e.g., Basu and Van (1998), Basu (1999), Beegle, R. H. Dehejia, and Gatti (2006), Beegle, R. Dehejia, and Gatti (2009), Bharadwaj, Lakdawala, and Li (2020), Edmonds (2005), and Udry (2006).

effects for mothers during the pandemic.¹² Descriptive evidence on the gendered impacts of COVID-19 in LMICs suggests that women increased domestic work and reduced their labor supply more than men, but causal estimates and analyses of the mechanisms behind labor supply changes are lacking.¹³ We show that time spent on childcare increased by about the same amount for both women and men in Kenya during the COVID-19 school closures in 2020, and returned to pre-pandemic levels after schools reopened in 2021. Contrary to evidence from other contexts, we find that the partial reopening of schools in October 2020 significantly increased adults’ labor supply with no significant difference by sex and that the increase was primarily driven by changes in child agricultural labor rather than childcare burdens.

The remainder of the paper is organized as follows. In Section 2 we describe the context around COVID-19 and school closures in Kenya, discuss the data used in this paper, and present descriptive statistics on childcare arrangements and child agricultural labor as school closures and other pandemic policies changed in Kenya. Section 3 presents the empirical approach. Section 4 presents overall results on the impacts of school reopenings on adult labor supply, while Section 5 explores mechanisms. Section 6 concludes.

2 Context and Data

This section summarizes Kenyan COVID-19 school closure policies, the data we use to analyze their impacts on labor supply, and information on childcare arrangements and child agricultural labor.

2.1 Kenyan School System and COVID-19 Closure Policies

Public primary and secondary education in Kenya is free for all children starting around age 6, and education is compulsory for the first nine years. Pre-school has also become broadly available, particularly for age 5 children. Academic years in Kenya begin in January and end in late October, and consist of three terms.

Schools in Kenya closed on March 16, 2020, days after the first reported COVID-19 cases, as part of a broad set of national restrictions to reduce the risk of disease transmission. The rest of academic Term 1 was cancelled. [Figure A1](#) shows a timeline of school closures and reopenings, other key pandemic-related policy changes, and weekly confirmed COVID-19 cases in Kenya.¹⁴ Pandemic school closure policy in Kenya was decided nationally, and is thus unrelated to local variation in economic or health conditions.

12. For example, Heggeness (2020) uses variation in the timing of school closures over space in the US and shows that mothers of school-age children with jobs were significantly more likely to not be working after schools closed, with no effects on working fathers and other women. Hansen, Sabia, and Schaller (2024) use variation in the timing of K-12 school reopenings over space in the US and show reopenings increased labor supply for married women with school-aged children only.

13. See e.g., Casale and Posel (2020), Chauhan (2021), Deshpande (2020), Grantham et al. (2021), Kugler et al. (2021), Ma, Sun, and Xue (2020), and Torres et al. (2023).

14. An overview of specific pandemic-related policies is presented in [Appendix B](#).

On September 15, the Ministry of Education released guidelines for safe reopening of schools, but the timing and nature of reopening remained uncertain until October 6, when the Ministry announced that students in grades 8 and 12—those sitting national exams—along with students in grade 4 should return to school on October 12 for Term 2 of 2020. This announcement was presented in the media as “a shocking move that caught parents and candidates off guard” ([The Star 2020](#)). On November 4, the President announced that schools would reopen fully for all students on January 4, 2021 to complete the 2020 school year. There were no additional fees incurred when schools reopened as parents had already paid fees for the 2020 school year, but some parents may have been asked to pay outstanding bills from before the school closures and others may have paid for new materials or extra lessons.¹⁵

Students in grades 4, 8, and 12 returned for Term 3 from January-March 2021 while all other students returned for Term 2; their Term 3 was shifted to May-July 2021. Grade 8 and 12 students sat national exams in March-April 2021. 2021 Term 1 for all students began in late July 2021. Terms and breaks for the 2021-2023 academic calendars were shortened to allow a gradual return to the standard pre-pandemic term schedule (running from January-October) in time for the 2024 academic year.

Kenya’s economy was beginning to recover from the initial COVID-19 shock by the time of the partial school reopening. Quarterly GDP growth was -4.1% and -3.6% in the second and third quarters of 2020, respectively, but 2.0% in the fourth quarter as the most restrictive pandemic policies had been lifted. GDP growth was 2.4% in the first quarter of 2021 and over 8.5% each quarter for the rest of the year. For comparison, quarterly growth rates for 2018-2019 and 2022-2023 were between 3.5-6% ([KNBS 2023](#)). Overall, Kenya’s economy was less adversely affected by COVID-19 than many other countries’, though effects were larger and the recovery more difficult for low-income populations ([FSD Kenya 2021](#)).

We focus on the impacts of the partial school reopening on adults’ labor supply for several reasons. First, unlike initial school closures, the partial reopening did not coincide with other pandemic-related policies (see [Appendix B](#)), allowing for cleaner identification. Second, we can exploit differences in the timing that children enrolled in different grades were eligible to return to school to isolate the effect of the shock. Third, analyzing impacts of availability of schooling for older children allows us to test both childcare and child labor mechanisms and understand how these affect adult labor supply in this setting.

2.2 Data

Data come from the Kenya COVID-19 Rapid Response Phone Survey (RRPS) panel, collected by the World Bank in collaboration with the Kenya National Bureau of Statistics and the University of California at Berkeley ([Pape 2021](#)).¹⁶ The main sample ($\sim 80\%$) is drawn from the nationally-

15. There are some additional costs associated with national exams, but these were paid by the government for all candidates at the time of the exams in Spring 2021.

16. See [Appendix C](#) for more detail.

representative Kenya Integrated Household Budget Survey conducted in 2015-2016, and this sample is supplemented by random digit dialing. The sample is intended to be representative of the population of Kenya using cell phones—80% of households nationally report owning a mobile phone, and these have better socioeconomic conditions on average than households that do not (Pape et al. 2021). We primarily use data from the first three survey rounds, covering May-November 2020, though we also use data from three rounds in 2021 to show the evolution of certain outcomes over time. In addition, we construct measures for certain variables in February 2020, before the first COVID-19 cases in Kenya, using recall questions from the first round.

The surveys include information on a rich set of socioeconomic characteristics. We use data on housing characteristics and ownership of major assets before the pandemic to construct a normalized wealth index, and categorize households as either above or below the mean of this index across the full RRPS sample. Household roster data includes the age and sex of all household members, as well as school enrollment information for all children. We describe how we use data on children’s grades at the time of the 2020 school closures to define the analysis sample in Section 3.

The outcomes of interest are measures of labor supply which are reported for all household adults (aged 18 or over).¹⁷ The extensive margin is measured by participation in the last 7 days in three activities: employed/wage labor, household non-farm enterprise, and household agriculture. The intensive margin is captured using hours of work by activity in the last 7 days; an individual not working in a given activity is coded as working 0 hours. The survey also includes data on total child hours spent working in household agriculture in the last 7 days, and whether the household hired any agricultural workers in this period. Child agricultural work hours are reported by 42.7% of agricultural households.

In addition to work hours, survey respondents also report their hours spent providing childcare hours in the last 7 days. This measure does not distinguish between time actively spent caring for a child and time spent on other activities while responsible for a child,¹⁸ and is unfortunately only available for other household adults starting in the September-November 2020 round and as a total for all household children starting in the January-March 2021 round. We topcode reported childcare hours at 140, or 20 hours a day.¹⁹

2.3 Childcare arrangements

At least 93% of children at each age from 6-16 in the RRPS are reported to have been enrolled in school in February 2020. In addition, over 90% of children age 5 were also enrolled in pre-school. Children primarily stayed at home with a parent during the COVID-19 school closures (Figure A2), including in situations where parents were simultaneously working. Almost no households report

17. We use the term labor ‘supply’ to refer to equilibrium outcomes, acknowledging that individuals may have been willing to supply additional labor but faced limited demand.

18. The survey explicitly asks respondents to consider time spent providing childcare alongside other activities, asking “In the past 7 days, how many hours have you spent doing childcare for your household, even if it overlapped with other tasks?”

19. This recognizes that, for young children especially, care responsibilities spill into sleeping hours. Results are qualitatively the same when topcoding childcare hours at 112, or 16 hours a day.

their children spending time with childcare providers outside the home or with a maid/domestic helper at home. In January-March 2021, 86% of households with children report 0 hours of care from non-household members in the last 7 days. This does not vary by rural/urban setting or change as pandemic restrictions were relaxed and case numbers fluctuated. While childcare availability has been increasing in Kenya (particularly in urban areas), affordability remains a challenge for most households (Clark et al. 2021; Murungi 2013). Such sources of childcare would therefore not have provided much or any relief to increased childcare burdens during the school closures period. Adults with schoolchildren at home will have faced trade-offs in their allocation of time across childcare, work in different sectors, and other activities given a limited time budget to accommodate increased childcare burdens.

Figure 1 Panel A presents how respondent childcare hours changed over time as school closure policy changed, separately for women and men. Female respondents provide around 15-20 more childcare hours than men outside the school closures period, but men still contribute around 25 hours on average. This contrasts with the image of fathers in African countries as primarily providing economic support and little childcare, but is consistent with recent evidence (Clark, Cotton, and Marteleto 2015; Kah 2012). We note, however, that the nature of care provided in these childcare hours might vary by adult sex, given the broad definition of childcare hours in the data. Childcare hours increased significantly by around 15-20 hours for both women and men during the school closures period.²⁰ Childcare hours began to drop in October-November 2020, coinciding with the partial reopening of schools,²¹ and returned almost exactly to pre-pandemic levels from the year before after schools fully reopened. These patterns suggest that the partial reopening of schools in October 2020 may have represented a decrease in childcare burdens for some households with children in grades 4 and 8.

To understand the conditions under which childcare needs would have been likely to fall, we consider data on childcare provision by all household members in the January-March 2021 survey round among analysis sample households (with at least one child in any grade from 3 to 9). This is after schools fully reopened but previous survey rounds do not include data childcare provided by children. Figure 1 Panel B shows how hours of childcare from different providers (excluding schools) vary with the number of household children. Respondents in the analysis sample provide 30-35 hours of childcare on average per week—46% of total household adult childcare hours, in line with most of these households having 2 adult parents²²— and this increases very little after the first child.²³ Households with more children may benefit from some economies of scale in childcare

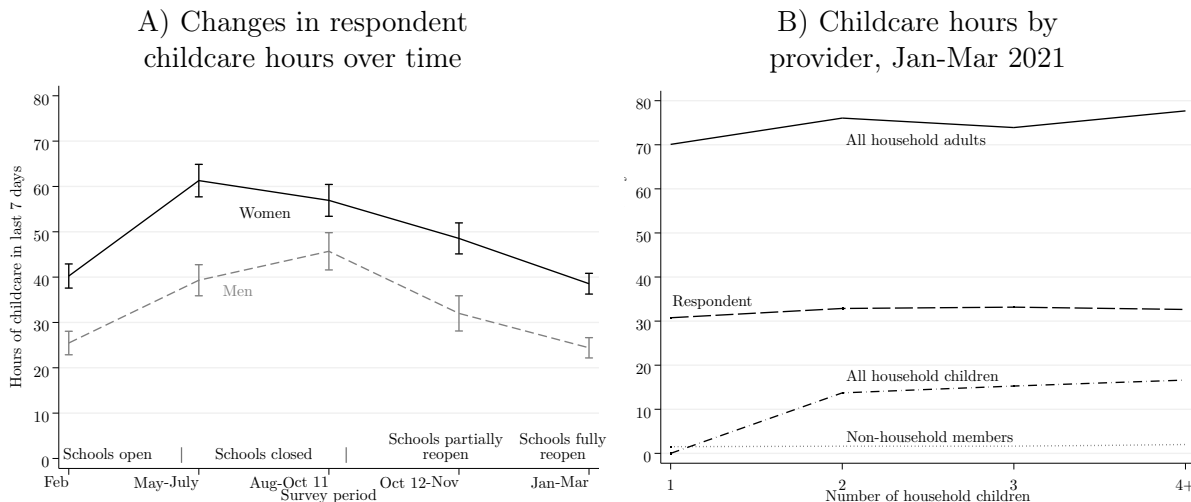
20. Other studies also find increases in domestic work during the pandemic for both men and women in India (Deshpande 2020), South Africa (Casale and Posel 2020), and many higher-income countries (see e.g., Andrew et al. 2020; Del Boca et al. 2020; Farré et al. 2020; İlkaracan and Memiş 2021), though most studies report larger increases for women.

21. In survey rounds 1 and 2 (May-September), respondents were asked to report only their own childcare hours. In survey rounds 3 and 4 (October-March), they were also asked to report childcare hours for other household adults. This change may explain part of the drop observed from August-October 11 to October 12-November, if it led respondents to reassess downward their own childcare hours in the context of total adult childcare hours.

22. Other adults besides the parents are present in 37% of households with children, and on average provide around 10 hours per week of childcare.

23. The survey measure of childcare includes ‘passive’ childcare when the adult is responsible for the child but not

Figure 1: Childcare hours in the last 7 days among analysis sample households, by provider of care and school closure status



Note: The figures show mean childcare hours in the last 7 days among analysis sample households (with at least one child in any grade from 3 to 9), by number of household children (ages 0-17) and by time period. Patterns are very similar when considering all households with at least one school-age child (age 5-17). Panel A presents data from RRPS round 4 (January-March 2021) which asks about childcare hours for each household adult, for all children in total, and for all non-household members in total. Previous rounds only ask about childcare hours for the respondent. The hours for 'all household adults' include the respondent's hours. Panel B presents data for female (black) and male (gray) *respondent* childcare hours over time as school closure policies changed. Data on childcare hours before and during the school closures period for other care providers are not available.

on the extensive margin, even if more children require more active attention during a given hour of childcare on the intensive margin. This result implies that the partial reopening will primarily have decreased childcare burdens in households where the child returning to school is the only child.

More generally, with additional children some will be older and thus require less care or supervision, and some will be net childcare providers. Consistent with prior studies in Kenya ([Jakiela et al. 2020](#)) and other LMICs ([Levison and Moe 1998](#); [Qureshi 2018](#)), siblings in the sample play an important role in childcare. In households with at least 2 children, 59% of children age 5-17—and 66% of children in grades 3-9—provided childcare to siblings in the last 7 days, for 15-20 hours on average in total.²⁴ The importance of sibling childcare in this setting suggests that a student returning to school might increase rather than decrease parents' childcare burden, in situations where they were net childcare providers while at home. This is most likely to be the case in households with young children.

actively providing care, meaning that adults responsible for their children for all (or most) of the day will report spending the same number of hours on childcare regardless of their number of children. But the median for reported childcare hours over the last 7 days is 20 hours in round 4, suggesting overly broad measurement is not solely responsible for the limited increase in childcare hours as the number of children increases.

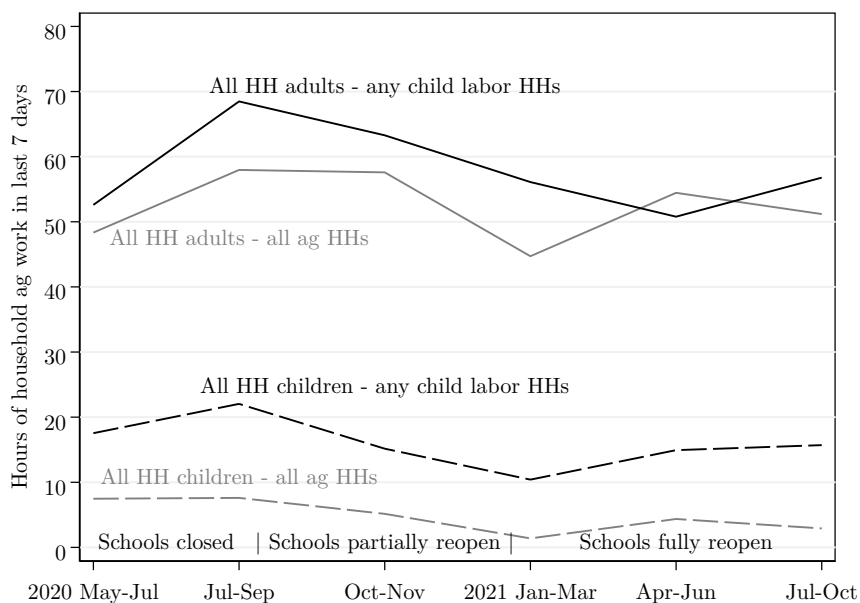
24. Sibling childcare provision as reported by a household adult may primarily represent 'active' childcare since that would be most easily observed and recalled by the respondent, and since they might report supervised sibling care-giving under their own childcare time rather than the child's. If this is the case, total hours of sibling childcare may be higher than what is reported in the survey.

2.4 Child agricultural labor

Moyi (2011) reports that around 30% of Kenyan children ages 6-14 engage in farm work (often alongside attending school), indicating this is fairly common in the study setting. In our analysis sample, 61% percent of households report having engaged in household agricultural work in the last 7 days during the school closures period. While we do not have data on individual child work engagement, 38% of agricultural households in the analysis sample report that children provided some farm labor.²⁵

Figure 2 shows how mean adult and child hours of work in household agriculture changed over time in the analysis sample. Adults in agricultural households work between 45-60 hours per week in all survey rounds, with some variation due to agricultural seasonality (September to November is the main harvest season for grains in most of Kenya). Children are reported to work less than 10 hours in all survey rounds on average across agricultural households. Differences in children’s agricultural work hours by round are clearer among households that report any child work, falling from an average of 20 hours in the last 7 days while schools were closed in 2020 to 15 hours during the same months in 2021 when schools were fully open. This 5 hour difference represents around 8% of average total household agricultural labor over this period.

Figure 2: Mean adult and child hours in household agriculture by survey round, agricultural households



Note: The figure shows mean reported hours of work in household agriculture over the last 7 days before the survey date for analysis sample households engaged in agriculture. The solid lines show total hours reported across all household adults—reported individually and then summed—while the dashed lines show total hours reported across all household children—reported as a single total by the respondent. The black lines include all agricultural households while the blue lines include only those reporting any child agricultural work hours during the round. Schools were fully closed during the first two survey rounds, partially reopened in October-November 2020, and fully reopened during the three 2021 survey rounds.

We note that child hours working in household agricultural are likely to be under-reported. First,

25. These numbers are nearly identical among the sub-sample of all households with any children ages 5-17. Child farm labor is not correlated with treatment prior to the partial reopening (Table A2).

social desirability bias may lead respondents to deliberately misreport child work hours. Second, respondents may struggle to accurately recall hours worked by all household children, since these are reported as a single number. The respondent is asked about household agriculture hours for each adult individual and then for the total across all children. In survey round 3, respondents report agricultural work hours for a randomly selected child as well as the total for all children. For households with a randomly selected child in grade 3-9 (in the analysis sample), 15% report more hours of work for the random child than they report for all children in total, more than the 10% that report more total child farm work hours; most of the agreement comes from reports of 0 hours of work. This indicates that estimates of child work on the household farm would likely be higher if respondents reported hours for each child individually rather than for all children in total.

Despite this measurement issue, it is clear that children can and do contribute to household farm work. In addition, the figure suggests that one effect of school closures may have been to increase children’s labor supply on the household farm. If households did not anticipate the partial school reopening in October 2020, which is likely given the late timing of its announcement, they may have had difficulties smoothing their labor supply. In particular, adults may have needed to increase their agricultural work hours to make up for a reduction in children’s work hours. In this case, we would expect larger increases in adult agricultural work hours in households where children were engaged in agricultural labor during the school closures period.

3 Empirical Approach

Our empirical strategy exploits quasi-random variation in which households are affected by the partial reopening, identified using information on what grades children were enrolled in at the start of the academic year interrupted by the pandemic school closures. Children in grades 4, 8, and 12 were eligible to return to school, and nearly 99% of eligible students are reported to have returned to school at the time of the partial reopening.²⁶

We compare ‘treated’ households with children enrolled in grades 4 or 8 prior to the pandemic (eligible for the partial reopening) against ‘control’ households with children in grades 3, 5, 6, 7, or 9, but not in grade 4 or 8. Few households report any children in grade 12, and we exclude these households from the analysis sample.²⁷ We separate ‘mixed’ households with children in both ‘treatment’ and ‘control’ grades from ‘treated’ households as they might experience different effects when not all children in the relevant grade range return to school. All three categories of households may also have additional children not enrolled in grades 3-9.²⁸ The main analysis sample includes

26. A survey of 3,000 grade 8 students in Busia County, Kenya similarly shows that 97% reported back to school after the partial reopening (Bonds 2021). Across all grades, 97% of previously enrolled students in the RRPS returned to school after the full reopening in January 2021.

27. Results are robust to including grade 12 students in the treatment definition and grade 10 and 11 students in the control definition, and to varying the grades included in the definition of control households (Table A8). All analysis sample households may also have other children outside of grades 3-9, which do not affect their treatment categorization.

28. 44% of analysis sample households have a child outside grades 3-9 and on average these households have 1.5 such children, with no difference by treatment group.

323 treated, 348 mixed, and 919 control households. These households are broadly similar to the full sample of survey households with any children (Table A1), but with 1 additional child on average, more engagement in household agriculture, and respondents more likely to be working (driven by agriculture).

We identify the effect of partial school reopenings through a difference-in-differences analysis comparing outcomes before and after the reopening between households with and without children who are eligible to return to school.²⁹ We estimate two-way fixed effects regressions of the form

$$y_{iht} = \alpha + \beta_1 \cdot Post_t \times Treated_h + \beta_2 \cdot Post_t \times Mixed_h + \mu_h + \tau_t + \epsilon_{iht} \quad (1)$$

where y_{iht} are outcomes for adults (age 18+) i in household h at time t , or household-level outcomes. $Post_t$ is an indicator for observations after the partial reopening on 12 October 2020.³⁰ We include observations from May-November 2020, omitting data from survey rounds after schools fully reopened. $Treated_h$ is an indicator for whether all household children in grades 3-9 were eligible to return to school. $Mixed_h$ is an indicator for whether the household has both eligible and ineligible children in this grade range. The omitted reference category is control households with children in this grade range but none eligible for the partial reopening. Household fixed effects μ_h absorb time invariant characteristics of households which may affect labor supply outcomes. Month fixed effects τ_t control for common shocks affecting households over time. We test robustness of the main results to specifications with different fixed effects and additional time-varying household- and individual-level controls and find that the coefficients generally remain stable (see Appendix E). We cluster standard errors at the household level.

Causal identification is based on the argument that unobserved factors that could affect labor supply outcomes are continuous around the thresholds of children being in adjacent grades, and did not differentially affect treated and control households around the time of schools reopening. As expected given the nature of the partial reopening, respondent and household characteristics are balanced across treated and control households during the school closures period (Table A2). Notably, we can reject differences in the pre-treatment school closures period for child engagement in household farm work, respondent childcare hours, and respondent engagement in different work activities.

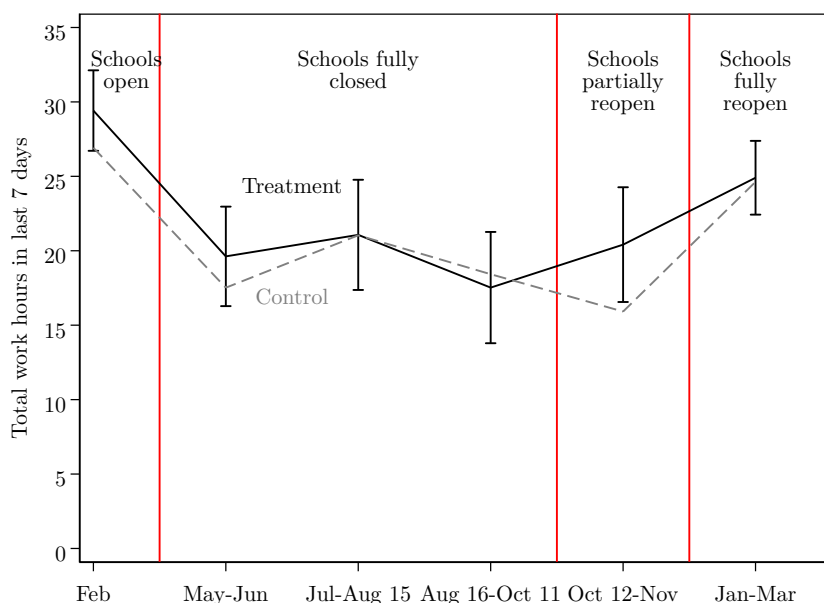
Critically for the identification strategy, Figure 3 shows that work hours trend almost identically

29. Since schools partially reopened all at once, our two-way fixed effects estimator should not suffer from the negative weighting issues that arise in staggered difference-in-difference designs (see e.g. Callaway and Sant’Anna 2021; Goodman-Bacon 2021).

30. The $Post$ coefficient is absorbed by the month fixed effects, where we include separate dummies for October 1-11 and 12-31. Survey round 3 began approximately 2 weeks prior to schools reopening, and 17% of analysis sample households were surveyed in round 3 before schools reopened. The order in which households were initially called was randomized in Round 1, and stayed the same in each subsequent round. Those 17% therefore represent a random subset of respondents, and are statistically similar to the analysis sample households surveyed after reopening (Table A5). Those surveyed earlier have younger respondents and are more likely to be urban and have a child age 0-4, perhaps because these households are easier to reach by phone. This limits our ability to generalize the results to the population of Kenya, but does not affect internal validity. These households do not inform the analysis and dropping them from the analysis sample does not affect our estimated coefficients (Table A8).

for respondents in treated and control households from February to early October 2020.³¹ Differences emerge following the partial reopening but are eliminated after schools fully reopen, when all households become ‘treated’. We find no significant differences between treated and control adults in overall labor supply in the periods when schools were fully closed, and cannot reject the joint hypothesis that the pre-reopening period treated coefficients are all 0 (Table A6). This indicates that treated and control households not only were on parallel trends prior to schools reopening—the minimum identification assumption required for our empirical strategy to be valid—but also statistically balanced in *levels*. There is also no evidence of anticipation effects in the period from August 16 to 11 October which is not surprising as the specific timing and the partial nature of reopening was not announced until the week before students were invited to return to school. Taken together, these results support the validity of the parallel trends assumption in our setting.

Figure 3: Mean respondent work hours by treatment group over time



Note: The figure shows mean total work hours in the last 7 days for survey respondents by treatment status over time, along with 95% confidence intervals around the treated group means. Means are shown for the respondent only due to missing data on February (pre-pandemic) working hours for other household adults. Data for February are based on recall from the first time a respondent is surveyed. Figure A5 shows differences in work hours by period for all adults in the analysis sample, excluding the February time period. Treated households have a child enrolled in grades 4 or 8, and control households have a child enrolled in grades 3, 5, 6, 7, or 9. Mixed households with children in both grade groups are not shown. The red bars indicate changes in Kenya’s school closures policy.

While adults in mixed households with children in both treated and adjacent grades have similar levels and trends in labor supply as control households during the school closures period, they differ significantly in certain characteristics which may affect the impact of the partial reopening (Table A2). Mixed households have 4.1 children on average compared to 2.9 in control households

31. We focus on respondents as work hours in February 2020 are available only for the respondent. Figure A5 shows differences in work hours by period for all adults in the analysis sample. The patterns are nearly identical though mean respondent work hours are higher than means for all adults, likely reflecting differences in age, household role, and reporting. The fall in hours in the period of the partial reopening for control households reflects the end of main harvest period in Kenya, as 64% of households are engaged in agriculture.

and 2.8 in treated households. This difference is mechanical, as mixed households must have at least one child in both the treatment and control grades.³² Respondents in mixed households are more likely to be married and have lower levels of education than control or treated households, indicating some differences correlated with having multiple children in this grade range. The difference in household composition and the fact that not all children in the target grade range in mixed households return to school leads to predictions of muted impacts of reopenings on this group of households. First, childcare hours are unlikely to change significantly when a similarly-aged child remains at home. Second, any decrease in child farm work hours may be partially offset by other children not returning to school.

4 Results

Table 1 presents results for the impacts of the partial school reopening on adults’ labor supply. Among control households, 56% of adults were working during the school closures period. Mean work hours of 15.7 in the last 7 days—27.6 among those that were working—reflect that many workers were not working ‘full-time’. Work is concentrated in household agriculture, which is common even in areas categorized as urban in the RRPS and would have been less affected by pandemic restrictions.

Table 1: Impacts of partial school reopening on adult labor supply

	N	Control Mean (SD)	Post x Treat (SE)	Post x Mixed (SE)
Engaged in any work in last 7 days	8694	0.557 (0.497)	0.073 (0.053)	0.063 (0.051)
Engaged in wage employment in last 7 days	8694	0.058 (0.234)	0.015 (0.012)	-0.006 (0.012)
Engaged in HH agriculture in last 7 days	8694	0.486 (0.500)	0.066 (0.055)	0.053 (0.049)
Engaged in HH non-ag enterprise in last 7 days	8694	0.067 (0.249)	0.004 (0.022)	0.015 (0.020)
Total work hours in last 7 days	8694	15.68 (19.68)	4.32** (1.88)	0.93 (1.79)
Wage employment hours in last 7 days	8694	1.91 (9.16)	0.61 (0.51)	-0.13 (0.51)
HH agriculture hours in last 7 days	8694	11.42 (15.29)	3.83** (1.56)	1.22 (1.51)
HH non-ag enterprise hours in last 7 days	8694	2.46 (10.81)	-0.07 (0.81)	-0.29 (0.76)

Note: This table presents estimates of Equation 1 for individual labor supply. Individuals not working in a given sector are coded as working 0 hours. From left to right, the columns show the dependent variable, number of observations, the control mean prior to the partial reopening, and the impacts of being in the partial reopening period for treated households (Post x Treat) and mixed households (Post x Mixed). Impacts for control households are absorbed by month fixed effects. Control households have a child in grades 3, 5, 6, 7, or 9, treated households have a child in grades 4 or 8, and mixed households have both. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. Standard errors are clustered at the household level. Data include observations for adults age 18+ from May to November 2020. Treated impacts on total and agricultural work hours remain marginally statistically significant after FDR multiple testing adjustment across all 8 Post x Treat estimates ($q = 0.086$ for both).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

32. Mixed households have 2.6 children in grades 3-9 on average, compared to 1.4 for control households and 1.2 for treated households, accounting for the entire difference.

We find no significant effects of the partial reopening on the extensive margin of labor supply. Though the point estimates represent an 11-13% increase in the probability of being engaged in any work in the last 7 days relative to control households, these are noisy ($p = 0.172$ for treated and $p = 0.213$ for mixed households). But adult labor supply responses to schools partly reopening for treated households are large and statistically significant on the intensive margin. Work hours in the last 7 days increase by 4.3 (27%) relative to adults in control households. This more than offsets a small reduction in hours worked in this time period for control households, potentially linked to the end of the main harvest season (Figure 3). ‘Mixed’ households with children eligible to return to school as well as children in adjacent grades do not change labor supply following the partial reopening. One child returning to school while another of a similar age stays home is unlikely to meaningfully reduce adult childcare burdens, and the child remaining at home may help make up for reductions in sibling childcare or agricultural labor provided by the child returning to school.

The increase in work hours for treated households is driven by increased engagement in household agriculture: a 3.8 (33.5%) increase in hours worked in the last 7 days ($p = 0.01$). Given low rates of engagement in non-agricultural activities, we lack statistical power to detect significant effects of a similar relative magnitude. The point estimate indicates a 31.9% increase in wage employment hours but this is not statistically significant ($p = 0.23$), and the estimated effect on household non-agriculture enterprise hours is close to 0. The increase in household agriculture hours rather than household enterprise may indicate that effects are driven by substitution for reduced labor after children who had been helping on the household farm return to school. On the other hand, positive point estimates for effects on wage employment participation and hours suggest reductions in childcare burden may also play a role. Null effects on non-ag enterprise may reflect that these activities were easier to combine with childcare. We discuss this further in the mechanisms section.

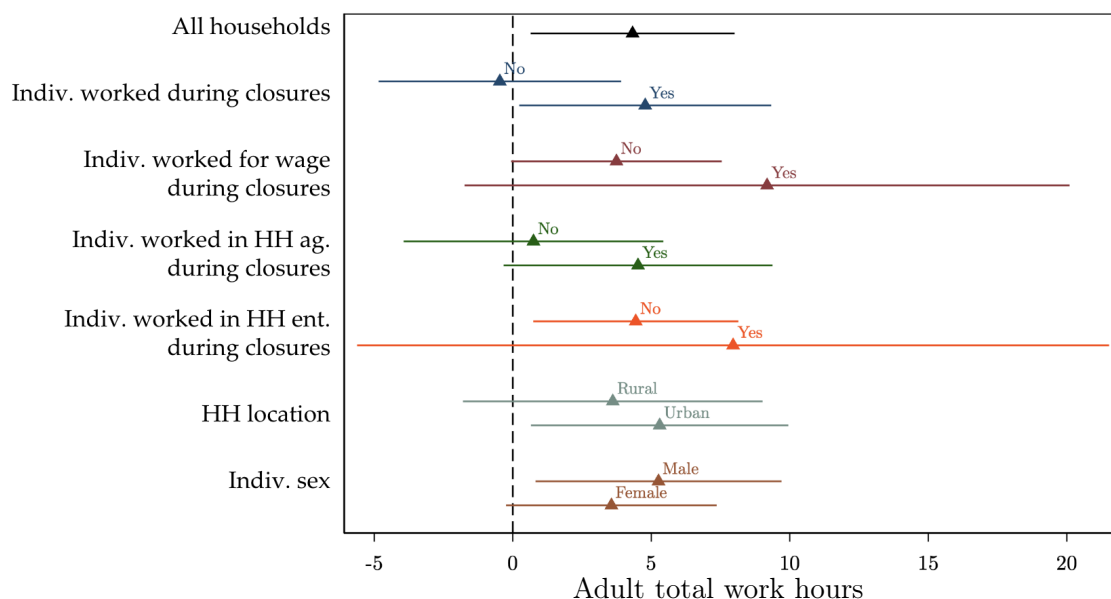
Greater impacts on household agricultural hours than in wage employment or household enterprise are not surprising given that we estimate short-term impacts in the weeks following the partial school reopening. Adults were more likely to pause their engagement in wage work and household non-ag enterprise due to pandemic restrictions. Engagement and hours in these activities may be constrained in the short run and take longer to adjust, while hours in household agriculture are more flexible.

We find evidence that adults adjust their labor supply in the short run on the most flexible margins by increasing hours in activities they were already engaged in, particularly in household agriculture (Figure 4). Treated adults that worked at any point while schools were closed (70%) increase their weekly work by 4.9 hours after one of their children returns to school, while those not working during the closures period do not significantly change their work hours after the partial reopening, with a point estimate close to 0 ($p = 0.04$ for the test of difference in effects).³³ Effects are driven by adults that worked in household agriculture during the school closures (63%). Despite

33. The increase in the impact of the partial reopening is larger across adults continuing work in all sectors, though the differences compared to those that did not continue working by sector are not statistically significant ($p = 0.12$ for household agriculture). This may be due to a lack of power, resulting from the small share of people working for either a wage (8%) or in a household enterprise (8%) during school closures.

the agriculture-driven effects, we observe no differences in impacts between urban (47%) and rural households. Part of this stems from the fact that the definition of ‘urban’ in the data includes many peri-urban areas; 43% of analysis households classified as urban are engaged in agriculture. Low take-up of formal childcare services and low wage employment in the sample may also limit urban/rural heterogeneity. We find no effect of the reopening in ‘large urban’ locations—those in counties with the largest cities in Kenya—consistent with impacts driven by agricultural work.

Figure 4: Heterogeneity in effects on adult work hours in the last 7 days by individual characteristics and household location



Note: The figure summarizes estimated coefficients and 95% confidence intervals for the effect on the partial reopening across individuals or households with different characteristics. Closures work participation is based on individual adult participation in a given sector from May-October 11 2020. Rural/urban location is defined by the RRPS. Observations include data from adults age 18+ in the analysis sample from May to November 2020. Results for mixed households are not shown. All regressions include household and month fixed effects, and SEs are clustered at household level. Full results are reported in [Table A12](#).

Impacts on work hours are not significantly different for women (54% of the sample) relative to men—men in treated households work 4.7 more hours after the partial reopening compared to 4.0 for women. This contrasts with evidence from high-income countries, which consistently shows larger effects of the pandemic on mothers’ labor supply relative to fathers’ and other women’s (e.g., Alon et al. (2022), Collins et al. (2021), and Hansen, Sabia, and Schaller (2024)). One explanation is suggested by the data on childcare hours in our context: responsibilities prior to the pandemic are less gendered than expected and both women’s and men’s hours increased by around 15 hours during school closures. So both women and men may be benefiting from a reduction in childcare hours. Another factor is that our study focuses on the return of a subset of household children rather than on effects of all children being either in or out of school, implying smaller changes in household childcare burden. Finally, slightly larger effects for men could reflect a greater importance of child farm labor as a mechanism behind these results. We analyze possible mechanisms further in Section 5.

In addition to effects of schools reopening on individual labor supply we also consider short-term effects on household-level earnings by activity over the last 14 days (Table A3). We see no statistically significant effects of the partial reopening on household income, and the point estimate for the impact on total household income is close to zero for treated households. Earnings data are limited—for all activities the 90th percentile of household earnings in the analysis sample is 0—in part due to a focus on the last 14 days, which does not accommodate seasonality or other variability in earnings, limiting our ability to detect impacts on income. Moreover, returns to agricultural labor hours may take until after harvest to materialize and the positive impacts of the partial reopening on agricultural work hours indicate treated households are still engaged in harvest activities which may explain the negative point estimate for agricultural earnings. The survey measures of income cannot be used to test whether these increased agricultural work hours led to increased incomes once harvest was completed.

We conduct a variety of robustness tests, focusing on the main impacts on total working hours in the last 7 days (Table A7, Table A8). Estimated impacts of the partial reopening on treated households are not sensitive to dropping month fixed effects, to the inclusion of individual or time-varying household controls, to using individual rather than household fixed effects, or to dropping household fixed effects and including household controls. Results are robust to defining *Post* by the date the potential reopening was announced rather than the actual reopening date. Slightly smaller estimated impacts in this specification together with no differential pre-trends suggest limited anticipation effects. Estimated effects on hours of work are not sensitive to focusing on sub-samples of adults more likely to be parent caregivers or engaged in work, though work hours increase by more for survey respondents. Finally, results are similar when dropping ‘mixed’ households and when varying which grades are included in the treatment group definitions.

5 Mechanisms

We now consider the mechanisms driving increased adult labor supply after a child returns to school. A simple model of household production, labor supply, and the intra-household allocation of different activities guides our analysis (we describe the model in more detail in Appendix D). Adults in the household supply labor to home production and other work activities, they provide childcare to children present in the household, and consume leisure. Adults get utility from consuming the returns to household labor, from leisure, and from altruistically caring about the well-being of their children.

Child well-being is subject to a constraint that total care received must not be below childcare needs, which vary by age and availability of school-based care. We assume adult childcare provision is decreasing in child age and exhibits increasing returns to scale as a function of the number of children receiving care in line with the patterns in Figure 1. Women may have advantages in childcare provision, for example due to social norms or particular needs of young children. Children present in the household are both recipients of childcare as well as potential contributors to home

production and of childcare to other siblings.

In the model, a child returning to school decreases the childcare demanded by that child while also reducing their provision of sibling childcare (if they have younger siblings), so the effect on the household childcare constraint depends on the returning child’s net childcare demand. A child returning to school also provides less household labor in households engaged in home production, which increases adult returns in home production relative to work outside the household. Although we do not model this explicitly, the returns to hiring workers from outside the household also increases.

Which effects dominate depend on the age distribution of household children and the economic activities the household engages in (which—following our main results—we consider fixed in the short run).³⁴

Separating households with children eligible to return to school into pure treated and mixed households implicitly tests for different effects by presence of additional adolescents in treated households. Table 1 shows no significant impacts of the partial reopening on mixed households. The presence of other adolescent children remaining in the household after schools reopen—demanding both similar levels of childcare and providing similar levels of sibling childcare and household productive labor as their siblings who return to school—mutes the potential mechanisms by which one adolescent returning to school could affect parent labor supply. Similarly, the lack of differences in effects on women’s and men’s work hours could imply that changes in childcare burdens are important for both, that both sexes substitute for reduced child agricultural labor, or that the mechanisms affect each sex differently but produce a similar total effect. We cannot conclude from the results thus far which mechanism is more important for the effects we observe in treated households.

The model predicts different effects of the partial reopening across households with different characteristics depending on whether changes in childcare burdens or child agricultural labor are a more important mechanism. We consider each of these in turn by testing for heterogeneity by household characteristics in impacts of the partial reopening on total adult work hours,³⁵ childcare hours for the respondent (not available for other adults), total household child agricultural work hours, and whether the household hired any agricultural workers from outside the household. For all tests, we interact the regressors in Equation 1 (except household fixed effects) with household or individual characteristics, corresponding to stacking separate regressions for each group. We focus the discussion on differences for treated households, as effects of the partial reopening for mixed households are not statistically significant.

34. In theory, school-related costs for a child returning to school may also lead to an income effect, increasing overall labor supply. But there were no additional fees incurred when schools reopened in Kenya. Though some parents may have paid outstanding bills and/or purchased school materials, this is unlikely to drive the labor supply effects we observe. To the extent there were costs of sending children back to school they would have been greater for households with children in private school (20% of households), but work hours do not increase (and the point estimate is negative) for households with any children in private school (Table A12). We therefore concentrate our discussion of mechanisms for the increase in treated households’ labor supply on changes in childcare needs and child labor.

35. These mirror effects on hours in household agriculture and we find no significant effects or heterogeneity for hours of work outside household agriculture (Table A17).

5.1 Childcare

We first test for evidence of a childcare mechanism. If changes in the childcare burden are the main mechanism behind the effects in Table 1, Figure 1 suggests we would expect small or null effects of the partial reopening on adult childcare hours and work hours if there are other children present in the household, and significant effects if there is only one child. The importance of sibling childcare suggests changes in childcare hours should be smaller or even negative if there are young children the older sibling might have been providing care for, implying smaller or null effects on adult work hours. To the extent girls were providing more childcare to siblings than boys while schools were closed—as suggested by evidence on gender differences in sibling childcare provision in similar settings (Jakiela et al. 2020; Levison and Moe 1998)—we would expect a girl returning to school to decrease childcare hours and increase adult work hours by less than a boy. As younger children are likely to demand more childcare than older children, decreases in childcare hours and increases in work hours should be larger in households where a grade 4 child returns to school than in households where a grade 8 child returns.

Figure 5 presents the tests of these predictions. Although the analysis of impacts on childcare hours is limited by the lack of data on adults other than the respondent and the broad definition of childcare in the survey, the results indicate that this mechanism is not a primary driver of increases in adult work hours after the partial school reopening.

We find no impact of the partial reopening on respondent childcare hours (Figure 5 Panel B). The point estimates are small and noisily estimated so we cannot reject increases or decreases of around 10 hours (roughly 20% of mean childcare hours during the school closures period).³⁶ Effects on respondent childcare hours do not differ significantly across any household characteristics,³⁷ though the signs of the differences by the presence of young children and a girl child are in line with predictions. The point estimates suggest respondent childcare hours decrease when there are no young children present, but *increase* when there are, particularly if the returning child is a girl. However, the differences by the presence of only 1 child or an older treated child are not in line with the childcare mechanism.

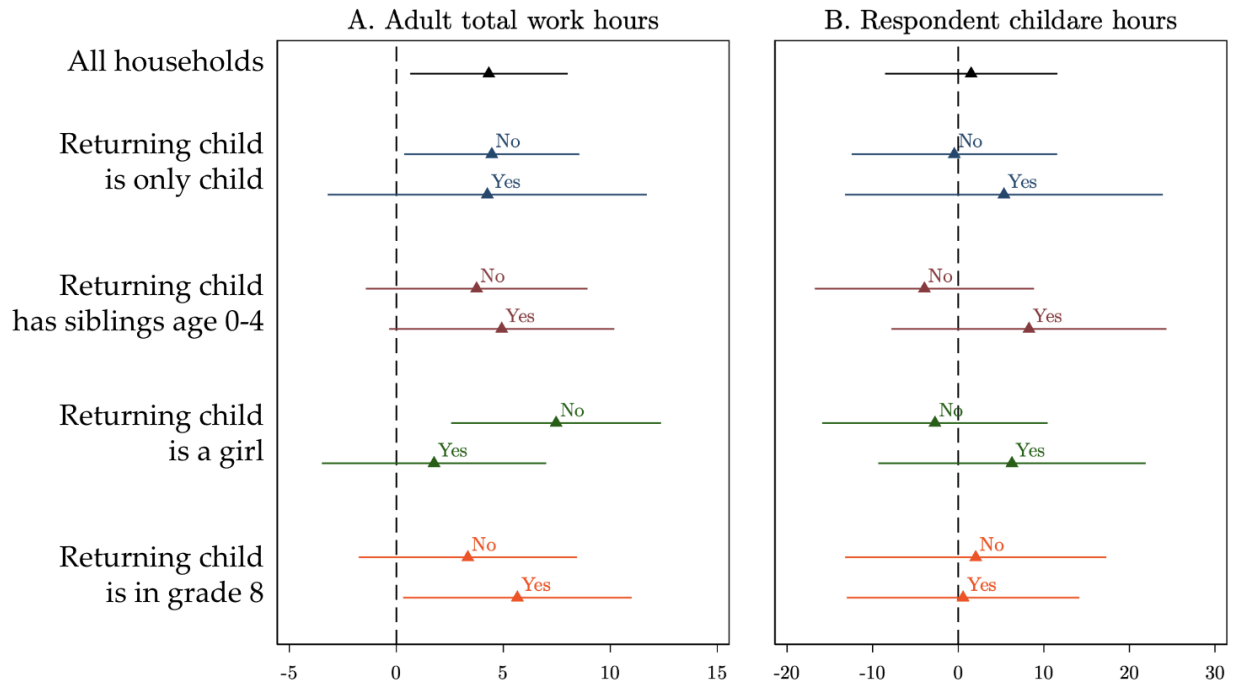
These imprecise estimates may be due in part to the broad definition of ‘childcare hours’ in the survey as including time spent caring for children while doing other tasks.³⁸ The data do not allow us to measure whether ‘active’ childcare hours—likely the main constraint to adult labor

36. Mean respondent childcare hours trend very similarly across treated and control households over the survey period for both women and men (Figure A3). Reported childcare hours fall by 10–15 for both treated and control households in the period after schools partially reopened. This may partly be due to a change in survey round 3 which began asking about childcare hours provided by other household members, potentially causing respondents to revise estimates of their own childcare hours downward.

37. We do not test for differences by respondent sex as households with female respondents differ from those with male respondents in a number of characteristics which could affect labor supply responses to the partial reopening. This is due to a survey coding error in the RRPS which led women to be requested as survey respondents whenever they were available in round 1. As respondent sex is balanced by treatment status (Table A2, this does not affect our identification strategy but prevents comparisons by respondent sex.

38. The survey measure of childcare hours can be thought of as a measure of the extensive margin of the childcare burden, as they combine time spent actively caring for children as well as passive time the adult is responsible for children but doing other activities.

Figure 5: Tests of childcare mechanism: heterogeneity in impacts of partial reopening on adult work hours and respondent childcare hours



Note: The figures show estimated coefficients and 95% confidence intervals for the effect of the partial reopening on adult work hours (Panel A) and respondent childcare hours (Panel B) across households with different characteristics. Outcomes are measured over the last 7 days. For control households, the ‘returning child’ is the child in a grade adjacent to those eligible to return. The comparison group for treated households where the returning child is a girl is control households with a girl in an adjacent grade. The comparison group for treated households where the returning child is in grade 8 is control households with a child in grade 7 or 9. Observations include data from adults age 18+ in the analysis sample from May to November 2020. Results for mixed households are not shown. All regressions include household and month fixed effects, and SEs are clustered at household level. Full results and tests of significant differences are reported in [Table A13](#) and [Table A14](#).

supply—were affected by the partial school reopening. Further, it is possible that changes in the respondent’s childcare hours do not accurately reflect changes for other household members. [Figure 1](#) shows that in round 4 respondents account for just under half of adult childcare hours. We observe both female (58%) and male respondents as well as non-household heads (36%), and importantly can show that the variation in reported adult childcare hours by count of household children is similar for respondents, household heads, and other household members ([Table A9](#)). This suggests that an analysis of respondent childcare hours should broadly reflect changes for other household adults. While we cannot rule out that the household childcare burden changes after the partial reopening, the results for respondent childcare hours therefore suggest no clear effects on adult childcare responsibilities.

In line with no changes in the childcare burden, we find no difference in effects of the partial reopening on total work hours by whether there is a young child in the household (47% of households) or by whether there is only one child (20%). Increases in adult work hours are larger in treated households where the child returning to school is a boy (45% of households). A boy returning to school increases work hours by 7.5 hours compared to 1.8 for a girl ($p=0.12$) but this difference is not reflected in effects on respondent childcare hours, suggesting a different mechanism. Looking

at effects by age of the treated child, work hours only increase significantly for households with a grade 8 child returning to school (54%) compared to when the child is in grade 4.³⁹ This is contrary to expectations for a childcare mechanism, but the point estimates are not significantly different: a 3.3 hour increase for a grade 4 child vs. 5.7 for a grade 8 child ($p=0.54$).

The patterns just described do not differ significantly between women and men overall (Table A10), indicating the childcare mechanism is not more relevant for women. One notable difference is that women but not men increase their hours of work outside of household agriculture by significantly more after the partial reopening in households with only 1 child (Table A11). This aligns with predictions based on a childcare burdens mechanism, where a child returning to school frees up time for adults to engage in less flexible work activities or activities away from the homestead. But this effect can only explain a small part of the average increases in work hours we observe in treated households after the partial reopening, which are driven by agricultural work.

In summary, while there are a few suggestive pieces of evidence on the childcare mechanism, other tests are inconsistent with its predictions. While childcare may have played a role, it cannot account for the overall adult labor supply increase.

5.2 Child labor

We next explore whether adult labor supply increases may be driven by the need to make up for reduced child labor. If so, we would expect effects on adult work hours, child engagement in household farm work, and the probability of hiring farm labor to be largest in households reporting child labor on the household farm during the periods when schools were closed. Effects should be larger in households with only one child, as other siblings are not available to help make up for reduced agricultural labor.⁴⁰ Finally, labor supply effects might be larger in less wealthy households which might be less able to hire outside labor to make up for reduced child labor. Figure 6 presents the tests of the child labor mechanism. The patterns in the data generally support predictions based on changes in child labor being the main mechanism behind the effects on labor supply we observe.

Average effects of the partial reopening on child labor hours in household agriculture are not statistically significant (the point estimate is -1; Table A15), but the decreases are larger in certain households (Panel B).⁴¹ Child agricultural hours fall by 3.7 hours in treated households with children

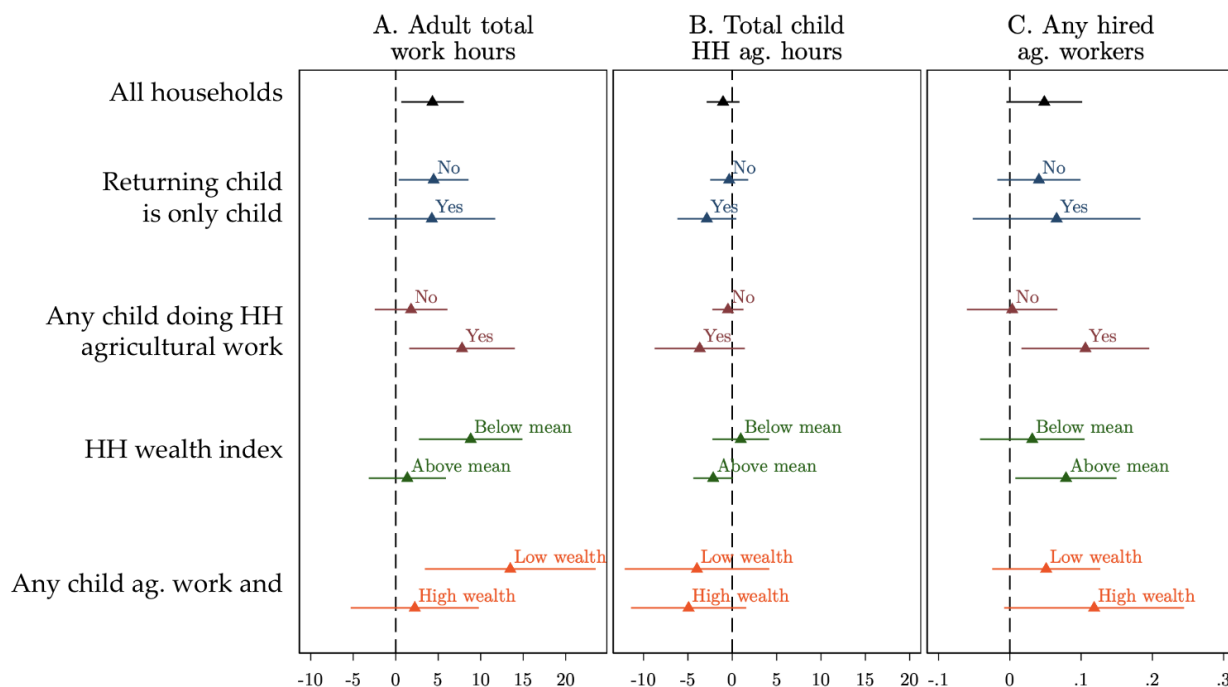
39. Among treated households, 56% have a grade 4 child and 54% have a grade 8 child. We do not separately consider those with children in both grades. The patterns are similar if we use a categorical treatment grade variable interacted with a dummy for ‘mixed’ over the full sample of analysis households.

40. The age and sex of the returning child might also be predicted to matter but we find limited differences in child work hours on these dimensions. Round 3 reports of individual child agricultural hours in the last 7 days for a randomly selected child show that children in grades 3-5 worked 2.3 hours compared to 3.5 for children in grades 7-9 and that boys in grades 3-9 worked 3.2 hours compared to 2.6 hours for girls.

41. Child agricultural work hours drop across all sample households over survey rounds 3 and 4 coinciding with the end of the main harvest season and before the start of the next planting season (Figure A4). Hours drop by more over the period of the partial reopening for treated households relative to control households—by 3.2 vs. 1.7 hours—and then are exactly equal in the period when schools are fully reopened. The patterns are similar when focusing on agricultural households, though for these the difference in the drop in child hours after schools partly reopen is larger: 7.3 hours for treated households compared to 3.6 for control households. Hours of work for individual children are not available before round 3, but reports for randomly selected children in round 3 after the partial reopening indicate

engaged in household agriculture during school closures ($p = 0.15$) and by 2.9 in those with only 1 child ($p = 0.09$), compared to point estimates very close to 0 in treated households without these characteristics. The estimates are imprecise and we cannot reject that the effects are the same across household types at conventional levels, but the results are consistent with predictions about where we would expect the largest effects. These decreases are not large in absolute terms but are large relative to the mean during the school closures period of 3.9 child agricultural work hours in the last 7 days.

Figure 6: Tests of child agricultural labor mechanism: heterogeneity in impacts of partial reopening on adult work hours, child agricultural work hours, and hiring any agricultural labor



Note: The figures show estimated coefficients and 95% confidence intervals for the effect of the partial reopening on adult work hours (Panel A), total child hours in HH agricultural work (Panel B), and the probability of hiring any agricultural labor (Panel C) across households with different characteristics. Outcomes are measured over the last 7 days. For control households, the ‘returning child’ is the child in a grade adjacent to those eligible to return. Child engagement in HH agricultural work is defined over the period when schools were fully closed. The HH wealth index is a normalized index based on housing quality and asset ownership prior to the pandemic. The bottom two rows restrict the sample households with children engaged in household agriculture and make comparisons by wealth. Observations include data from adults age 18+ in the analysis sample from May to November 2020. Results for mixed households are not shown. All regressions include household and month fixed effects, and SEs are clustered at household level. Full results and tests of significant differences are reported in [Table A13](#), [Table A15](#), and [Table A16](#).

Larger decreases in child agricultural work hours in households where they are engaged in household agriculture (41%) align with the much larger increases in adult work hours in these households (Panel A). Adults in treated households with children working in agriculture work 7.8 more hours after the partial reopening relative to control households, while adults in other treated households do not significantly increase their work hours. Patterns of heterogeneity in effects on total work hours are similar for both women and men ([Table A10](#)), indicating both sexes are affected by this

differences of similar magnitude: 1.9 hours of work for children in grades 4 or 8 and 2.6 for children in adjacent grades among all analysis sample households, and 2.7 vs. 5.0 among those reporting any child agricultural work.

mechanism.

The magnitudes of the impacts of the partial reopening on adult work hours and child agricultural work hours in households with child labor do not quite align, as the decrease in child hours of 3.7 is much smaller than the increase of 7.8 hours per adult. This may reflect under-reporting of child agricultural work due to either social desirability bias or data collection issues as described in Section 2.4. However, we have no reason to expect different reporting by treatment status: baseline child engagement in household farm work is balanced (Table A2) and hours reported for individual children in round 4 after schools fully reopened do not differ by treatment status. Potential measurement issues should therefore not bias the direction of the estimated effect of the partial reopening on child agricultural work hours, but limit our ability to confidently estimate the magnitude of the effect. The difference in magnitudes may also imply that another mechanism explains part of the effect of the partial reopening, including potential changes in childcare burdens that we cannot observe with the data we have available.

Households reporting child agricultural labor differ from other sample households in many characteristics besides being more agricultural which could potentially explain this heterogeneity, such as being less wealthy (Table A4), but the data suggest the decrease in child work hours is the most likely explanation. In particular, among less wealthy households only those reporting child labor during school closures see adult work hours increase, while those without child labor do not. These results imply adults are increasing agricultural work to substitute for reduced work hours by the child returning to school. Smaller decreases in child agricultural work hours in households with more than 1 child indicate that other household children also help to make up for the lost work hours of the returning child. As adults do not reduce childcare hours or non-agricultural work hours, these increases in work hours appear to be drawn from time that had been allocated to leisure or other domestic activities.

Panel C shows effects on the probability of hiring agricultural labor. On average, the partial reopening increases the probability that treated households hire any agricultural labor in the past 7 days by 0.05 ($p = 0.08$), a 70% increase relative to the control mean (Table A16). This change is driven by an increase of 0.11 in households using child agricultural labor compared to 0 in households that do not ($p = 0.07$ for the difference), implying some households are substituting child agricultural labor with hired labor. This aligns with findings from Allen (2024) in Malawi that overlap between the harvest period and the school calendar increases household expenditures on hired labor.

The child labor mechanism also reconciles the difference in effects on adult work hours and hired labor by household wealth, proxied by a normalized index based on housing quality and asset ownership prior to the pandemic.⁴² Adults in households with below mean wealth (40%) increase

42. Household wealth is positively correlated with being in an urban location, electricity grid connection, and engagement in household enterprise, and negatively correlated with the number of household children, being engaged in household agriculture, and having children working in agriculture (Table A4). Less wealthy agricultural households are also less likely to hire agricultural labor than wealthier households: hired labor was reported by 3% and 8% of these households during the school closures period, respectively.

work hours by 8.8 hours after the partial reopening compared to 1.4 hours for households with higher wealth ($p=0.05$). In line with the child labor mechanism, this increase in work hours for adults in low-wealth households is driven entirely by low-wealth households reporting child labor during school closures (Panel A, bottom two rows). Poorer households, however, are not significantly more likely to hire outside workers (Panel C, bottom four rows). In contrast, adults in high-wealth households where children engaged in agricultural work during school closures do not report working more hours (Panel A, bottom row). Instead, these households are 11.8 percentage points ($p=0.07$) more likely to hire additional workers, and explain almost the entire average increase in hired labor (Panel C, bottom row). These results indicate that adults in wealthier households partly substitute child labor with hired labor rather than their own labor, while less wealthy households may lack this option and therefore increase adult labor.

In short, the evidence above is consistent with child labor in household agriculture being an important feature of our context, and the main mechanism behind the increases in adult work hours following the partial school reopening, particularly among poorer and agricultural households.

6 Conclusion

We present nationally-representative estimates of the impacts of a shock to childcare burdens and child labor on adult labor supply in a lower-middle-income African country using pandemic-related school closure policy changes in Kenya as exogenous childcare shocks. We find large and positive impacts of Kenya’s partial school reopening on adult labor supply for households with a child eligible to return, but limited evidence of effects through changes in the household childcare burden. The limited childcare effects lead to similar impacts of schools reopening on both women and men, in contrast to studies of pandemic school closures in high-income countries. This does not indicate that childcare is not important for parents’ labor supply in Kenya but rather that this particular policy did not meaningfully change childcare burdens. The return of a single child to school from households which nearly all have multiple children does not significantly affect respondent childcare hours, either by reducing time spent caring for that child or making up for time that child spent caring for siblings, and effects on adult work hours do not vary in ways that would be expected under a childcare mechanism.

Instead, the evidence indicates that increased adult work hours after the partial reopening, which coincides with the end of the main harvest season in most of Kenya, are driven by reductions in child labor in household agriculture. Patterns of reductions in child labor align with increases in adult work hours with greater changes in treated households where children worked in agriculture during school closures, particularly in less wealthy households. More wealthy treated households are more likely to hire outside labor after the partial reopening, further evidence of a need to make up for reduced child labor. These results are consistent with child agricultural labor being an important input for many households in low-income countries, with implications on the relative return to schooling.

The household labor supply shock we analyze takes place in the context of a global pandemic, but the results on labor supply impacts will continue to have relevance. Further school closures may be enacted in response to future outbreaks of COVID-19 variants or other diseases despite strong evidence of adverse effects on children. In addition, although some pandemic-related restrictions were still in effect at the time schools partly reopened in Kenya in October 2020, many (including the most severe) had been relaxed even if economic activities had not yet fully returned to normal. The impacts we estimate may therefore generalize to policies or events affecting household childcare needs and child labor supply in similar settings, such as mandatory secondary schooling or reduced school costs. They also shed light on potential effects on adult labor supply of school closures for reasons unrelated to public health events, such as teacher strikes (e.g., Jaume and Willén 2021 in Argentina)—schools in Kenya were closed due to teacher strikes in 2013, 2015, and 2024 (BBC News 2013, 2015; Atiena 2024).

While school closures in Kenya affected the availability of child labor for households, school breaks more generally determine when and how much children are available to work. Changes in the Kenyan academic calendar following the COVID-19 school closures varied the timing of school breaks relative to periods of peak agricultural labor demand each year from 2020 to 2023. The current calendar includes a break between terms 1 and 2 around the time of the main planting season in most of Kenya, but term 3 overlaps with much of the main harvest season. Our results suggest that such overlap may significantly affect labor availability on household farms, and the opportunity costs of children attending school for less wealthy households in particular. This is in line with recent work that has studied how the overlap between school days and peak agricultural periods in developing countries affects children’s school participation and advancement (Allen 2024) and children’s labor supply (Merfeld 2024). Future work could consider how differences in this overlap affect adults’ labor supply, use of hired farm labor, and agricultural productivity, as well as human capital formation and longer-term effects for children in agricultural communities more broadly.

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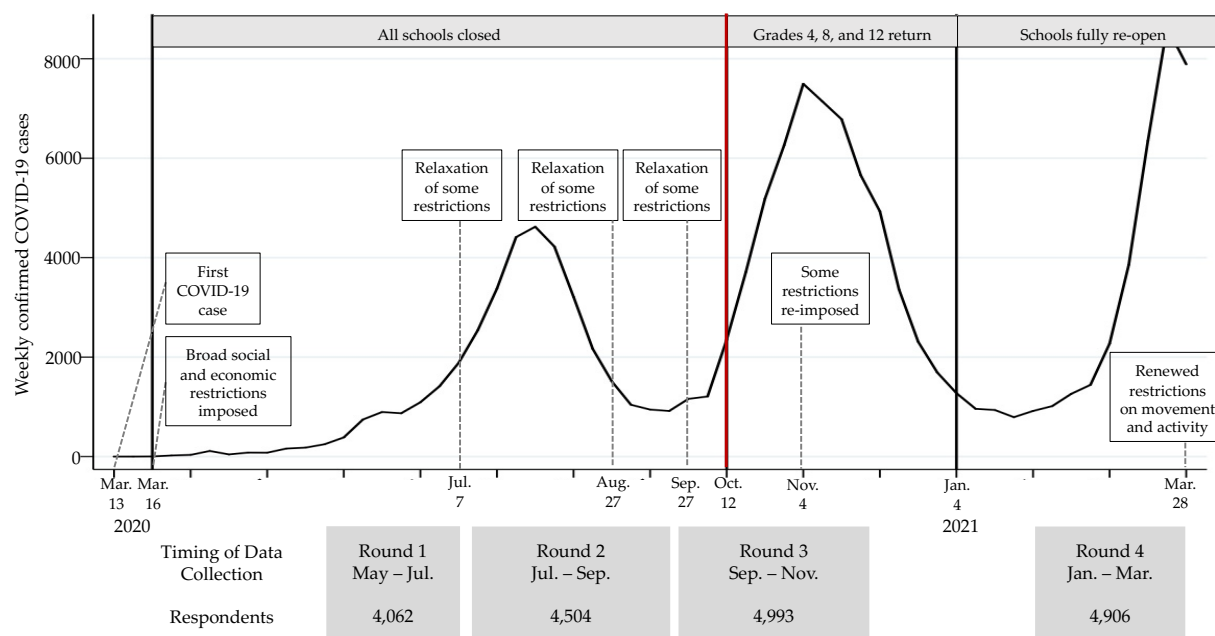
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Appendix A: Additional Figures and Tables

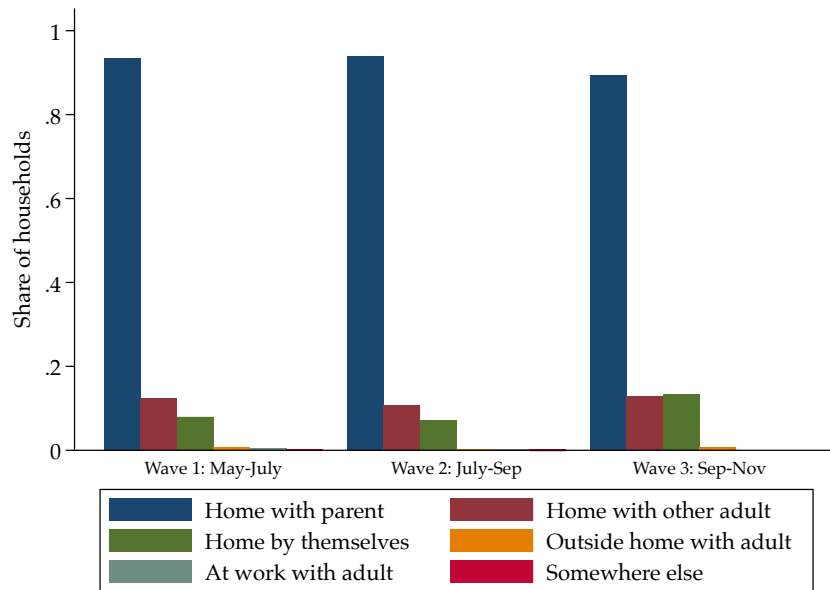
Figure A1: Kenya COVID-19 cases, pandemic policy, and data collection timeline



Note: The figure shows the evolution of weekly confirmed COVID-19 cases in Kenya over time, along with the timing of key pandemic policy changes. The red bar indicates the partial school reopening on 12 October, the focus of the analysis. 'Relaxation of some restrictions' indicates that one or more of the initial pandemic constraints were at least partially reduced. Specific policy changes are outlined in [Appendix C](#).

Sources: [COVID-19 government response timeline for Kenya](#); [Kenya COVID Tracker](#); [Presidency of Kenya](#); [Kenya Ministry of Education Twitter feed](#); [COVID-19 Data Repository by the Center for Systems Science and Engineering \(CSSE\) at Johns Hopkins University](#)

Figure A2: Childcare arrangements when children are out of school



Note: Respondents are asked to specify all of the situations where a randomly selected child spent at least some time when out of school in the past week. 'Somewhere else' combines 'daycare/other childcare' and 'at home with a maid/domestic helper.' The figure uses information on childcare arrangements for analysis sample households with at least one child in any grade from 3 to 9, but the distribution is nearly identical when considering all households with children.

Table A1: Comparison of analysis sample to other survey households

Sample Observations	(1) All households 7171	(2) Households with children 4901	(3) Analysis households 1590	(4) All HHs ever using child ag labor 1494	(5) Analysis HHs ever using child ag labor 631
<i>Household characteristics</i>					
Female household head	0.32 (0.47)	0.30 (0.46)	0.29 (0.45)	0.28 (0.45)	0.26 (0.44)
Age of household head	43.48 (14.35)	43.35 (13.11)	44.81 (12.01)	48.08 (13.04)	47.01 (12.33)
Count adults	2.40 (1.28)	2.61 (1.27)	2.61 (1.21)	2.77 (1.32)	2.70 (1.19)
Count of all kids age 0-17	1.44 (1.72)	1.98 (1.75)	3.12 (1.60)	2.43 (1.97)	3.44 (1.73)
Any child age 0-4	0.25 (0.44)	0.35 (0.48)	0.37 (0.48)	0.28 (0.45)	0.35 (0.48)
Household wealth index	0.05 (1.01)	0.02 (0.99)	-0.06 (0.94)	-0.22 (0.93)	-0.27 (0.89)
Connected to electricity grid	0.54 (0.50)	0.50 (0.50)	0.46 (0.50)	0.35 (0.48)	0.31 (0.46)
Urban	0.53 (0.50)	0.51 (0.50)	0.47 (0.50)	0.42 (0.49)	0.38 (0.49)
HH engaged in agriculture in last 7 days	0.48 (0.50)	0.55 (0.50)	0.61 (0.49)	0.87 (0.34)	0.88 (0.33)
Any child HH ag work in last 7 days	0.15 (0.35)	0.21 (0.41)	0.27 (0.44)	0.66 (0.47)	0.64 (0.48)
HH engaged in non-ag enterprise in last 7 days	0.14 (0.35)	0.16 (0.36)	0.16 (0.36)	0.12 (0.32)	0.11 (0.32)
<i>Respondent characteristics</i>					
Age	39.39 (13.81)	38.66 (12.42)	40.56 (11.64)	42.53 (13.36)	42.38 (12.30)
Female	0.52 (0.50)	0.56 (0.50)	0.58 (0.49)	0.53 (0.50)	0.54 (0.50)
Completed primary school	0.89 (0.32)	0.89 (0.32)	0.87 (0.34)	0.83 (0.38)	0.82 (0.39)
Completed secondary school	0.55 (0.50)	0.52 (0.50)	0.47 (0.50)	0.42 (0.49)	0.37 (0.48)
Completed school beyond secondary	0.21 (0.40)	0.19 (0.39)	0.15 (0.35)	0.13 (0.33)	0.11 (0.31)
Married	0.64 (0.48)	0.72 (0.45)	0.75 (0.43)	0.74 (0.44)	0.78 (0.42)
Is the household head	0.71 (0.45)	0.66 (0.47)	0.64 (0.48)	0.64 (0.48)	0.64 (0.48)
Engaged in any work in last 7 days	0.59 (0.49)	0.63 (0.48)	0.68 (0.47)	0.84 (0.37)	0.84 (0.37)
Engaged in wage employment in last 7 days	0.13 (0.34)	0.12 (0.32)	0.10 (0.30)	0.07 (0.26)	0.06 (0.24)
Engaged in HH agriculture in last 7 days	0.43 (0.49)	0.49 (0.50)	0.56 (0.50)	0.80 (0.40)	0.82 (0.39)
Engaged in HH non-ag enterprise in last 7 days	0.09 (0.29)	0.10 (0.30)	0.10 (0.30)	0.07 (0.26)	0.07 (0.26)
Engaged in any work in February 2020	0.74 (0.44)	0.77 (0.42)	0.83 (0.37)	0.84 (0.36)	0.90 (0.30)

Note: The table compares means for household and respondent characteristics across different samples of survey households. Data are from the first time a household is observed, typically in survey round 1 (May-early July) while schools were fully closed. Column (1) includes all households surveyed at least once in the first 4 rounds of the survey. Column (2) includes households with any children. Column (3) includes analysis sample households: those with at least one child in grades 3-9. Column (4) includes all households reporting any child HH ag work at any point during the first 4 rounds of the survey. Column (5) is the same but restricted to analysis sample households. "HH" indicates "household".

Table A2: Baseline balance by treatment status

	Control		Treatment		Mixed		C-T	C-M
	HH N	Mean	HH N	Mean	HH N	Mean	p-value	p-value
<i>Household characteristics</i>								
Female household head	919	0.29	323	0.30	348	0.26	0.826	0.208
Age of household head	919	44.20	323	46.07	348	45.27	0.018	0.149
Count adults	919	2.55	323	2.65	348	2.74	0.223	0.016
Count of all kids age 0-17	919	2.89	323	2.77	348	4.06	0.213	0.000
Any child age 0-4	919	0.38	323	0.34	348	0.39	0.305	0.753
Household wealth index	919	-0.05	323	0.03	348	-0.14	0.147	0.132
Connected to electricity grid	919	0.46	323	0.50	348	0.41	0.171	0.136
Urban	919	0.46	323	0.48	348	0.47	0.587	0.701
HH engaged in agriculture in last 7 days	919	0.61	323	0.59	348	0.64	0.575	0.254
Any child HH ag work in last 7 days	919	0.25	323	0.25	348	0.32	0.804	0.026
HH engaged in non-ag enterprise in last 7 days	919	0.15	323	0.19	348	0.15	0.071	0.811
<i>Respondent characteristics</i>								
Age	919	39.97	323	41.34	348	41.38	0.081	0.051
Female	919	0.59	323	0.56	348	0.57	0.328	0.654
Completed primary school	916	0.88	323	0.87	348	0.84	0.619	0.086
Completed secondary school	916	0.48	323	0.47	348	0.41	0.811	0.030
Completed school beyond secondary	916	0.15	323	0.16	348	0.13	0.482	0.483
Married	908	0.74	316	0.72	343	0.81	0.520	0.013
Is the household head	919	0.64	323	0.66	348	0.63	0.592	0.729
<i>Respondent labor supply</i>								
Engaged in any work in last 7 days	919	0.68	323	0.69	348	0.67	0.554	0.834
Engaged in wage employment in last 7 days	919	0.10	323	0.12	348	0.08	0.329	0.197
Engaged in HH agriculture in last 7 days	919	0.55	323	0.53	348	0.59	0.512	0.184
Engaged in HH non-ag enterprise in last 7 days	919	0.09	323	0.13	348	0.10	0.059	0.690
Engaged in any work in February 2020	919	0.82	323	0.86	348	0.84	0.101	0.401
Childcare hours in last 7 days	917	48.36	322	45.31	348	51.10	0.284	0.346

Note: The table presents means for control households (C) with a child in grade 3, 5, 6, 7, or 9, treated households (T) with a child in grade 4 or 8, and mixed households (M) with a child in both grade groups. Data are from the first time a household is observed, typically in survey round 1 (May-early July) while schools were fully closed. Individual-level data are for the survey respondent. Columns on the right present p -values for tests of equality for control households compared to treated and mixed households, separately. The joint F-stat for differences across control and treated households is 0.94, with p -value 0.543. It is 8.45 ($p < 0.001$) for differences across control and mixed households.

Table A3: Impacts of partial school reopening on household income in the last 14 days

	KSH				Log(KSH+1)			
	(1) Total	(2) Wage work	(3) HH ag.	(4) HH ent.	(5) Total	(6) Wage work	(7) HH ag.	(8) HH ent.
Post x Treat	-122.4 (656.9)	287.9 (466.1)	-259.5 (176.7)	-150.8 (428.5)	-0.600 (0.369)	-0.176 (0.241)	-0.308 (0.257)	-0.209 (0.227)
Post x Mixed	-813.8 (688.2)	-99.6 (404.6)	-299.0** (138.5)	-415.2 (534.1)	-0.570 (0.362)	-0.202 (0.237)	-0.365* (0.214)	-0.123 (0.230)
Observations	2997	2997	2997	2997	2997	2997	2997	2997
Mean, pre-reopen control	2404.0	833.8	303.8	1266.3	2.145	0.708	0.735	0.894

Note: This table presents estimates of [Equation 1](#) for measures of household income over the last 14 days. Columns 1-4 report estimates for income in Kenyan Shillings (KSH; USD 1 $\approx$ 107 KSH) and columns 5-8 report estimates for the log of income + 1. Observations include data from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by 'Treat'), control households with children in an adjacent grade, and 'mixed' households with both types of children. 'Post' is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. SEs clustered at household level.

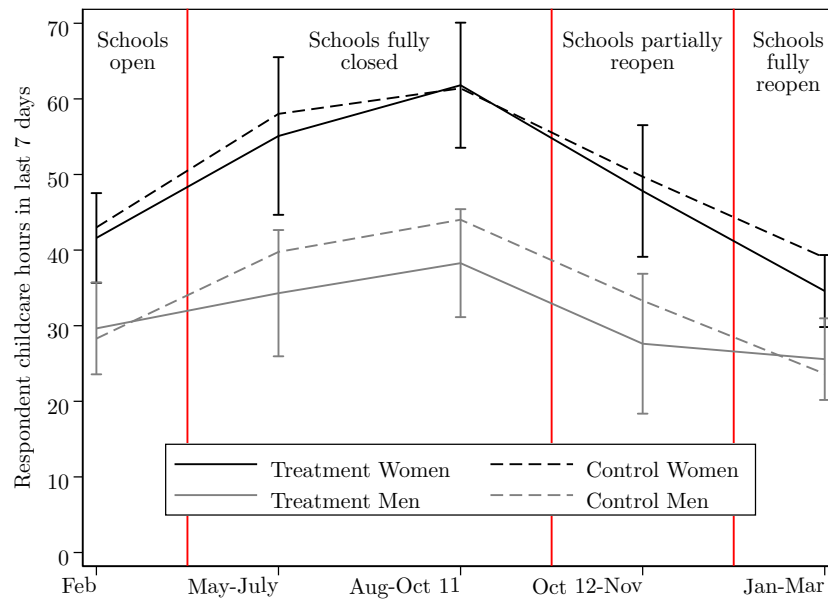
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Correlates of sample household characteristics

	(1) Above mean HH wealth	(2) Any child HH ag work
Treat	-0.012 (0.029)	0.016 (0.028)
Mixed	-0.008 (0.030)	0.050* (0.028)
Female household head	-0.009 (0.026)	-0.046* (0.025)
Age of household head	-0.002** (0.001)	0.003*** (0.001)
Count adults	-0.007 (0.010)	-0.023** (0.010)
Count of all kids age 0-17	-0.018** (0.009)	0.042*** (0.008)
Any child age 0-4	-0.023 (0.027)	-0.100*** (0.025)
Connected to electricity grid	0.326*** (0.024)	-0.098*** (0.024)
Urban	0.046** (0.023)	-0.013 (0.022)
Household engaged in agriculture	-0.063** (0.027)	0.392*** (0.024)
Any child ag labor	-0.045* (0.026)	
Household engaged in enterprise	0.088*** (0.031)	-0.022 (0.030)
Above mean household wealth		-0.040* (0.024)
Observations	1590	1590
Outcome mean	0.590	0.397

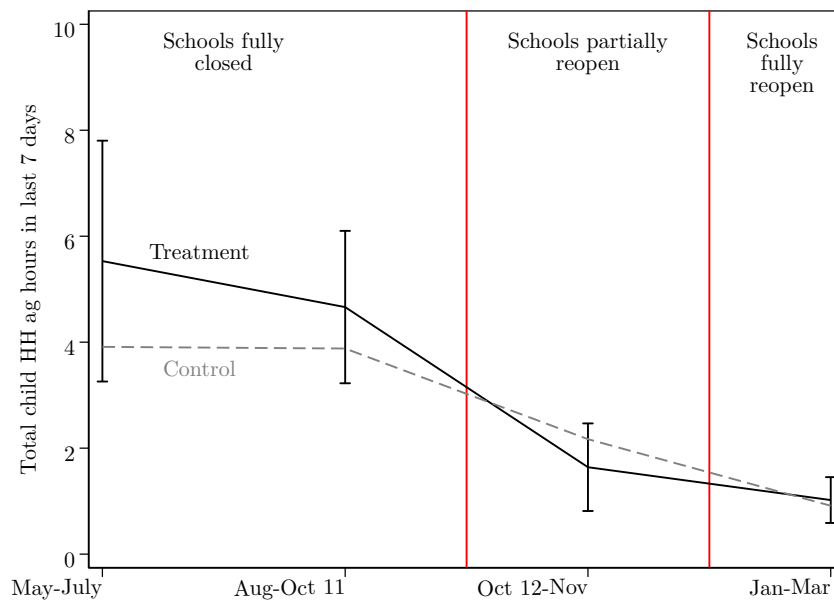
Note: This table presents correlates of whether the household is above the mean for a normalized index of household wealth (based on housing and asset ownership before the pandemic) and whether the household reports any children engaged in household agricultural work. Data are from the first time a household is observed in the RRPS. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A3: Mean respondent childcare hours by treatment group and sex over time



Note: The figure shows mean respondent childcare hours in the last 7 days by treatment status and sex over time, along with 95% confidence intervals around the treated group means. Childcare hours include time spent doing other activities while caring for children. Treated households have a child enrolled in grades 4 or 8, and control households have a child enrolled in grades 3, 5, 6, 7, or 9. Mixed households with children in both grade groups are not shown. The red bars indicate changes in Kenya's school closures policy.

Figure A4: Mean total child hours in household agricultural work by treatment group over time



Note: The figure shows mean total child hours in household agriculture in the last 7 days by treatment status over time, along with 95% confidence intervals around the treated group means. Hours are reported in aggregate for all children rather than individually by child. Treated households have a child enrolled in grades 4 or 8, and control households have a child enrolled in grades 3, 5, 6, 7, or 9. Mixed households with children in both grade groups are not shown. The red bars indicate changes in Kenya's school closures policy.

Appendix B: Major Pandemic Policy Changes in Kenya

The following outline summarizes when major nation-wide pandemic-related policies were implemented and relaxed over the course of 2020 after the first COVID-19 cases in Kenya on March 13. The dates for the announcements of new restrictive policies are in *italics* and the dates when these policies were relaxed or ended are in **bold**. We also include announcements related to school closures, even though policies did not necessarily change with these announcements. Most policies were extended multiple times after first being imposed; we do not list the dates of policy extensions, except for school closures.

- *March 13-20*
 - Suspend all public gatherings, meetings, games, events
 - Ban on gatherings of more than 10 people
 - All schools closed
 - Recommend working from home where possible
 - Ban on foreigner entry; quarantine requirements for entry of nationals and visa holders
 - Public transport asked to reduce to 60% of capacity
- *March 24-27*
 - Ban on national and international flights
 - Closure of bars and restaurants for in-person service
 - Direct cash payments implemented for vulnerable citizens
 - Stay at home requirements imposed, except for ‘essential’ trips
 - Curfew imposed from 1700 to 0500 hours
 - Public transit closed between ‘infected’ and ‘not infected’ areas
- April 26: School closures extended to June 4
- **April 27**: Partial reopening of restaurants for take-out service
- June 6: School closures extended until further guidance from the Ministry of Health
- **June 7**: Nightly curfew revised to between 2100 and 0400 hours
- June 24: Announcement that school might reopen on September 1
- **July 7**
 - Phased reopening of religious gatherings
 - Up to 100 people permitted to attend weddings and funerals
 - Local air travel within Kenya to resume July 15
 - International air travel to resume August 1
- July 7: Announcement that schools will remain closed until January 2021, final exams are cancelled, and students would repeat the year; colleges and universities following strict guidelines might reopen in September
- *July 27*
 - Restaurants reopened, must close by 1900 hours
 - Ban on sale of alcoholic drinks and beverages in eateries and restaurants
- **August 27**
 - Restaurants may remain open until 2000 hours
 - Ban on sale of secondhand clothing lifted
 - Licensed hotels may sell alcohol
- September 15: Ministry of Education releases guidelines for safe reopening of schools

- September 21: Ministry of Education calls all teachers to report back to schools by September 28
- **September 27**
 - Nightly curfew revised to between 2300 and 0400 hours
 - Bars may reopen; restaurants and eateries may sell alcohol; bars, restaurants, and eateries may remain open until 2200 hours
 - Religious gatherings may open for up to 1/3 of capacity
 - Up to 200 people may attend funerals and weddings
- October 6: Ministry of Education announces that students in examination grades (4, 8, and 12) shall return to classes on October 12
- **October 12:** Students in examination grades (4, 8, and 12) to return to classes
- *November 4*
 - Requests for government work to be done remotely when possible
 - Political gatherings suspended
 - Nightly curfew revised to between 2200 and 0400 hours
 - Bars, restaurants, and eateries must close by 2100 hours
- November 4: Announcement that schools to fully reopen in January 2021
- **January 4:** Schools fully reopen

Other policies were implemented that specifically affected certain parts of the country. For example, on April 6 the government instituted a 21 day movement ban/lockdown for Nairobi, Kilifi, Kwale, and Mombasa, and Mandera was added soon after. This lockdown was extended multiple times. These were the only counties affected. The lockdowns for Kilifi and Kwale ended on June 7 and those for Nairobi, Mombasa, and Mandera ended on July 8.

Sources: [COVID-19 government response timeline for Kenya](#); [Kenya COVID Tracker](#); [Presidency of Kenya](#); [Kenya Ministry of Education Twitter feed](#)

Appendix C: Data

Data come from the Kenya COVID-19 Rapid Response Phone Surveys (RRPS), collected by the Kenya National Bureau of Statistics with support from the World Bank and the University of California at Berkeley. [Pape et al. \(2021\)](#) describe the survey methodology and implementation in detail.

The main RRPS sample is drawn from the nationally representative Kenya Integrated Household Budget Survey (KIHBS) conducted in 2015-2016: 9,009 households that were interviewed and provided a phone number served as the primary sampling frame for the RRPS. All households in the sample were targeted in each round regardless of whether they were reached in a previous round. By the fourth round of the RRPS, 5,499 KIHBS households had been successfully surveyed at least once. The KIHBS sample is supplemented by random digit dialing (RDD). From a sampling frame of 5,000 randomly selected numbers, of which 4,075 were active, 1,554 households had completed at least one survey by round four.

The sample is intended to be representative of the population of Kenya using cell phones. In the 2019 Kenya Continuous Household Survey 80% of households nationally report owning a mobile phone, though certain counties—notably in the northeast—have much lower mobile phone penetration. [Pape et al. \(2021\)](#) report that KIHBS households that provided a phone number and those that were successfully surveyed in the RRPS have better socioeconomic conditions—measured by housing materials and asset ownership—than households that did not provide a phone number or that did but were not reached for the RRPS. The RRPS data include household survey weights adjusting for selection and differential response rates across counties and rural/urban strata, attempting to recover national representativeness. We do not apply these household weights for our individual-level regression analyses.

We primarily use data from the first four rounds of the RRPS, covering May 2020-March 2021, and also construct measures for February 2020, before the first COVID-19 cases in Kenya using recall questions from the first time a household was surveyed. We also show data on household agricultural labor for two later rounds of the RRPS covering April-October 2021 for comparison with reports during the same months in 2020. Each round lasted approximately 2.5 months and covered a representative cross-section of households each week within each round. The order in which households were called was randomized in the first survey round and maintained in all subsequent rounds.

Due to a survey coding error, in round 1 women were requested as survey respondents whenever they were available leading to a higher share of female respondents. Subsequent rounds were conducted with the same respondent unless they could not be reached. The survey team attempted to reach the target respondent at least 5 times across several days and hours of the day before moving on to speak to any other knowledgeable household member in order to avoid replacing the respondent. Due to this non-random selection of respondents, household characteristics differ by whether the baseline respondent is female or male.

The surveys include information on household composition, assets and housing, labor outcomes for household adults, and child schooling and care, as well as more general household information and COVID-specific modules. Detailed questions on child care, schooling, labor, and other outcomes are included for a randomly selected child in each round.

Data on childcare arrangements for a randomly selected child include questions on which household member has primary responsibility for the child’s care, which household member was with the child in the last 15 minutes, and where and in whose company the child stayed during the day

when out of school (from a set of general categories).⁴³ The surveys also ask respondents for their hours spent on childcare in the last 7 days.⁴⁴ Childcare hours from other household adults are measured starting in round 3, while childcare hours from all household children combined and all non-household members combined are included starting in round 4, after schools fully reopened.

In the first survey round, respondents are asked about their housing characteristics and ownership of selected assets before March 2020. We consider whether the dwelling has a primary floor material other than earth, dung, or palm and a primary wall material of cement, stone, bricks, adobe, or wood planks, and whether the household owned a radio, mattress, refrigerator, and mobile phone (the four most commonly reported assets). We normalize each of these components, sum them, and normalize the sum across the entire survey sample to construct our normalized wealth index.

Labor supply is captured using modules on household agricultural production, household enterprise, and wage employment. For both household agricultural production and for each household enterprise, respondents report all household adults engaged in those activities over the last 7 days and their hours of work. Wage employment is reported for each household adult. An individual not working in a given activity is coded as working 0 hours. Recall on labor supply for February 2020 is only available for survey respondents. Households also report whether they hired any outside workers to work on household agricultural activities.

Respondents report estimates of total household earnings from agricultural activities and from household enterprises over the last 14 days. For the few households with multiple enterprises, we sum earnings and profits across enterprises. Wage earnings in the last 14 days are reported for individual wage workers. For comparability with the measures of household agriculture and enterprise earnings, we aggregate wage earnings to the household level. Earnings data are limited—for all activities the 90th percentile of household earnings in the analysis sample is 0—in part due to a focus on the last 14 days, which does not accommodate seasonality or other variability in earnings.

We winsorize reported individual hours of work and household earnings across activities at the 99th percentile. We winsorize reported childcare hours at 140 per week.

43. Respondents are instructed to select all childcare arrangements used. Nevertheless, respondents might omit types of childcare that are used less frequently or that are seen as less socially acceptable (e.g., leaving a child at home by themselves).

44. The survey asks “In the last 7 days, how many hours did you spend doing childcare?” and does not distinguish between time actively spent caring for a child and time spent on other activities while responsible for a child. We topcode reported childcare hours at 140, or 20 hours a day. Over 15% of respondents in our analysis sample indicate spending at least this many hours on childcare.

Appendix D: A simple static partial equilibrium model of childcare and labor supply

Here, we develop a simple model of adult labor supply and childcare decisions to generate predictions to take to the data. The model considers a static problem for adult household members with children. For simplicity, we assume that household adults take decisions jointly, and thus model the decision as that of a single person. We focus on a static, partial equilibrium labor supply decision, and set aside possible impacts of shocks to labor demand to focus on shocks to the adolescent school availability. Key aspects of the context that we aim to reflect in the model are the availability of child labor in household agriculture as well as childcare of younger children by older children. To reflect women's larger role in childcare in this setting, we assume that female adult household members have either a comparative advantage in childcare of younger children or, similarly, that social norms are such that the costs of refraining from childcare for women (or the cost of engaging in childcare for men) are exogenously larger. Moreover, in line with our data, we assume economies of scale in childcare provision, as well as the ability of the caregiver to combine some types of childcare with household production in agriculture.

Household adults get utility $U = U(C, L, \{Q_k\}_{k=1}^N)$ from consumption, leisure, and the well-being of k household children. They can spend time on leisure, wage work for a fixed wage, home production⁴⁵, and childcare and face a time constraint $T = L + t_w + t_h + t_c$. Wage work earns a wage w . Home production $H = H(t_h, X, S)$ is a concave function satisfying the Inada conditions, which depends on the adult time input and other household characteristics X such as the availability of household agricultural land or existence of a household enterprise, as well as the availability of child labor (which is a function of the age distribution of children and school closure policy S). We normalize the price of consumption to 1 and assume that household production can be sold at this same price.

Adults provide childcare $CCM = \psi_g(t_c + \theta_h t_h)$, which includes “active” childcare time t_c focused on children as well as some portion θ of home production time that simultaneously provides passive childcare. ψ_g is a cost-shifter for childcare provision that takes on smaller values for men than for women.⁴⁶ Adult childcare is a public good that all household children can access, reflecting the economies of scale we observe in this context.

Total childcare for child k is given by $CC_k = CCM + I_k$, adults' childcare provision plus any childcare provided by older children.⁴⁷ All children receive the same amount of care from household adults, but childcare from other children varies based on the age distribution of siblings. Total informal childcare available to the household $I = I(X, S)$ is a function of household characteristics (notably the presence of older siblings) and of school closures; older siblings provide more informal childcare when schools are closed. Child well-being $Q_k = Q_k(CC_k, \bar{C}C_k(\text{age}_k, S))$ is a concave function of childcare provided to the child and their minimum required care.⁴⁸ Minimum required care $\bar{C}C_k(\text{age}_k, S)$ decreases with age, and for school-age children it increases when schools are closed.

45. The key distinction this model makes is between a work sector which accommodates both simultaneous childcare and child labor contribution, and another work sector which does not. We therefore primarily think of the home production activity as being household agriculture, with household enterprise activities being a form of home production which has some of the same characteristics but less so.

46. This cost-shifter can be rationalized in multiple ways. It could be that women's childcare hours count for more relative to men's due to a social expectation of women to provide childcare, or a social stigma of childcare for men. This shifter is also isomorphic to a model where women are relatively more productive in childcare, and require fewer hours to achieve the same increase in child welfare.

47. Very few households in the data use non-household sources of childcare, so we abstract away from this possibility.

48. We can think of this as a Stone-Geary type of function.

Adults take as given household characteristics X such as presence of other adults and child composition, school closure policy S , and non-labor income Y . We model S as a binary variable taking a value of 1 if schools are closed and 0 otherwise.⁴⁹

Adults' static optimization problem is

$$\max_{t_w, t_h, t_c} U(C, L, \{Q_k\}_{k=1}^N) \quad (2)$$

Subject to

$$C = wt_w + H(t_h, X, S) + Y \quad (3)$$

$$T = L + t_w + t_h + t_c \quad (4)$$

$$L \geq 0; t_w \geq 0; t_h \geq 0; t_c \geq 0 \quad (5)$$

$$CC_k = \psi_g(t_c + \theta_h t_h) + I_k \quad (6)$$

$$\sum_{k=1}^N I_k \leq I(X, S) \quad (7)$$

$$Q_k = Q_k(CC_k, \bar{C}C_k(\text{age}_k, S)) \quad (8)$$

Adults maximize their utility over choices of time use, subject to the following constraints: 1) the household budget constraint, 2) their time endowment, 3) non-negativity constraints on time use, 4) the childcare provision function, 5) availability of childcare from older siblings, and 6) the child well-being function. The budget constraint states that spending on consumption must equal the sum of wage income, the value of home production, and non-labor income.

We are interested in the impacts of changes in school closure policy S on adult labor supply t_w and t_h . S enters the model through childcare needs, the availability of sibling childcare, and household child labor availability. We expect $\bar{C}C_k(\text{age}_k, 1) > \bar{C}C_k(\text{age}_k, 0)$ for children enrolled in school, meaning schools being open decreases household childcare needs. On the other hand, $I(X, 1) > I(X, 0)$ for households with older children, as those children can provide more informal childcare when schools are closed. Moreover, $H(t_h, X, 1) > H(t_h, X, 0)$ as children may contribute more to home production when they are home from school.

Adults thus trade off their time among wage work, home production, childcare, and leisure, at an interior solution setting the marginal returns to each equal to each other:

$$u'_C w = u'_C H'_t(X, S) + \phi(\{\text{age}_k\}_{k=1}^N, S, \psi_g) \theta_h = \phi(\{\text{age}_k\}_{k=1}^N, S, \psi_g) = u'_L \quad (9)$$

where these terms are, respectively, the marginal utility of working one more hour for a wage w , the marginal utility of an additional hour in home production (providing both consumption and some child well-being value due to joint work and childcare), the marginal value to adults of an additional hour of childcare, and the marginal utility of an additional hour of leisure, and where we define

$$\phi = \sum_{k=1}^N u'_{Q_k} Q'_k(\text{age}_k, S) \psi_g \quad (10)$$

School reopening (moving from $S = 1$ to $S = 0$) affects the solution to adults' problem through two channels. First, it lowers child labor in home production, thus likely raising the marginal return of adults working in home production after school reopenings ($H'_t(X, 0) > H'_t(X, 1)$). Second,

49. S may vary by child, as in the case of the partial school reopening in Kenya, but we abstract from this point.

it changes the net demand (and thus net return) to childcare for the remaining children. When there are no other children present, the utility return to adult childcare hours decreases for sure ($\phi(K = 1, S = 1) < \phi(K = 1, S = 0)$), which also reduces the return to work in home production. But when there are other children present, the sign of this effect is ambiguous and depends on the age distribution of remaining children. When remaining children are very young, for instance, the marginal utility from adult childcare may actually *increase*, since the school reopening decreases sibling childcare ($I(X, 1) > I(X, 0)$) and changes the household childcare constraint.

The simple model thus allows us to generate several hypotheses for the effect of schools reopening to take to the data. First, in households where *all* children return to school, hours spent in childcare should decline while labor supply overall should increase. The decline in childcare should be greater for a younger child returning to school. Second, adults with young children will likely be providing a higher level of childcare on average when schools are open: sibling childcare falls as older siblings return to school, but childcare needs remain high since these are driven relatively more by younger children. Sibling childcare may be particularly affected by a girl child returning to school. Labor supply may therefore decrease in households with young children given the increased marginal utility from adult childcare.

Third, if women play a larger role in childcare (as is the case in our setting), either because children benefit more from female care, or because social norms and economic circumstances are such that men's social cost to childcare are larger, then $\psi_{female} > \psi_{male}$. This suggests that women should engage in more childcare overall, and that school closures should impact their childcare hours more. They will also be more likely to supply relatively more labor to home production, as this can be combined with childcare whereas wage work cannot. If, in addition, we suppose that childcare needs and norms for infants and young children are particularly gendered, then labor supply responses to school reopening when young children remain present in the household should be particularly muted (or even reversed) for women relative to men.

Finally, in households which used child labor for home production while schools were closed, adult labor supply in home production should increase after schools reopen to make up for lost child labor. This increase should be more pronounced in households which depend more on home production and which lack access to hired labor, such as poorer agricultural households. Households with better access to hired labor may respond to reduced child labor by increasing hired outside agricultural labor rather than household adult agricultural labor, if the opportunity cost of household adult labor is greater than the cost of hired labor.

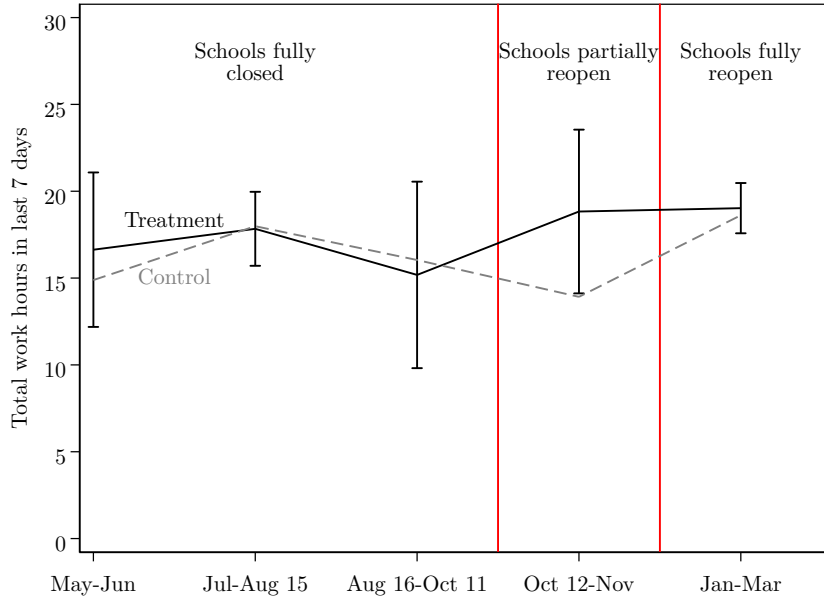
Appendix E: Robustness

Table A5: Baseline balance by timing of round 3 interview

	Post-Oct 12		Pre-Oct 12		p-value,
	HH N	Mean	HH N	Mean	diff=0
<i>Household characteristics</i>					
Female household head	1386	0.28	204	0.30	0.584
Age of household head	1386	44.92	204	44.08	0.346
Count adults	1386	2.61	204	2.64	0.755
Count of all kids age 0-17	1386	3.10	204	3.27	0.140
Any child age 0-4	1386	0.36	204	0.47	0.004
Household wealth index	1386	-0.04	204	-0.16	0.107
Connected to electricity grid	1386	0.46	204	0.46	0.921
Urban	1386	0.45	204	0.57	0.001
HH engaged in agriculture in last 7 days	1386	0.61	204	0.59	0.599
Any child HH ag work in last 7 days	1386	0.27	204	0.26	0.929
HH engaged in non-ag enterprise in last 7 days	1386	0.15	204	0.19	0.184
<i>Respondent characteristics</i>					
Age	1386	40.87	204	38.42	0.005
Female	1386	0.58	204	0.61	0.373
Completed primary school	1384	0.87	203	0.88	0.652
Completed secondary school	1384	0.46	203	0.47	0.810
Completed school beyond secondary	1384	0.14	203	0.17	0.411
Married	1364	0.76	203	0.72	0.312
Is the household head	1386	0.65	204	0.58	0.075
<i>Respondent labor participation</i>					
Engaged in any work in last 7 days	1386	0.68	204	0.65	0.323
Engaged in wage employment in last 7 days	1386	0.11	204	0.08	0.297
Engaged in HH agriculture in last 7 days	1386	0.56	204	0.55	0.965
Engaged in HH non-ag enterprise in last 7 days	1386	0.10	204	0.09	0.531
Engaged in any work in February 2020	1386	0.83	204	0.83	0.938
Childcare hours, last 7 days	1383	47.65	204	53.07	0.133

Note: The table presents means for households surveyed during RRPS round 3 (late September-November) after schools partially reopened on October 12 or before. Data are from the first time a household is observed, typically in survey round 1 (May-early July) while schools were fully closed. Individual-level data are for the survey respondent. The column on the right presents p -values for tests of equality between these groups of households, separately. The joint F-stat for differences across household groups is 2.10, with p -value 0.001.

Figure A5: Mean adult work hours by treatment group over time



Note: The figure shows mean total work hours in the last 7 days for adults in control households over time (dashed line), and in each period we add to this the estimated treatment effect with 95% confidence intervals using the results from Table A6. For the reference time period in that regression we show the treated group mean and 95% confidence interval. Data on February (pre-pandemic) working hours are not shown as they are only available for the respondent. Treated households have a child enrolled in grades 4 or 8, and control households have a child enrolled in grades 3, 5, 6, 7, or 9. Mixed households with children in both grade groups are not shown. The red bars indicate changes in Kenya's school closures policy. We fail to reject that the treated differences in the periods while schools are closed are jointly 0 ($p = 0.58$).

Table A6: Impact of partial reopening on adult labor supply in the last 7 days, by time period

	(1) Engaged in any work in last 7 days	(2) Total work hours in last 7 days
Treat × May-Jun	0.02 (0.06)	1.75 (2.27)
Treat × Aug 16-Oct 11	-0.05 (0.07)	-0.86 (2.74)
Treat × Oct 12-Nov	0.07 (0.07)	4.91** (2.40)
Mixed × May-Jun	0.04 (0.06)	0.55 (2.28)
Mixed × Aug 16-Oct 11	0.10 (0.07)	2.54 (2.74)
Mixed × Oct 12-Nov	0.11* (0.06)	1.83 (2.31)
Observations	8694	8694
Mean, pre-reopen control	0.59	16.21
p -value, pre-reopening coefs jointly 0 for treatment	0.58	0.49

Note: This table presents estimates from Figure A5 of the interaction between *Treat* and time period from Equation 1, where *Post* is replaced with time period dummies, which also enter separately into the equation. The reference period is July-August 15, while schools were closed and before the partial reopening was announced. We show p -values from the F-test that the coefficients for May-June and Aug-Oct 11 are jointly 0 for treated households. Observations include data for all household adults (age 18+) from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by 'Treat'), control households with children in an adjacent grade, and 'Mixed' households with both types of children. Regressions include household and month fixed effects. SEs clustered at household level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Impacts of partial school reopening on total working hours, varying controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post		-3.28*** (0.95)				-2.26** (0.88)	0.97 (0.65)
Post x Treat	4.32** (1.88)	4.39** (1.89)	4.31** (1.88)	4.19*** (1.42)	4.41** (1.89)	2.34 (1.85)	2.81** (1.38)
Post x Mixed	0.93 (1.79)	0.92 (1.81)	0.92 (1.79)	0.29 (1.36)	0.95 (1.80)	0.97 (1.68)	-0.23 (1.23)
Observations	8694	8694	8694	8694	7898	7898	7898
Mean, pre-reopen control	16.21	16.21	16.21	16.21	16.05	16.05	16.05
Household FE	Y	Y	Y	Y	N	N	N
Individual FE	N	N	N	N	Y	N	N
Month FE	Y	N	Y	Y	Y	N	N
Individual controls	N	N	Y	Y	N	N	Y
Household controls	N	N	N	Y	N	N	Y

Note: This table presents estimates of Equation 1 with varying controls. The dependent variable is total working hours over the last 7 days, taking a value of 0 for individuals not working. Observations include data from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by ‘Treat’), control households with children in an adjacent grade, and ‘Mixed’ households with both. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Column 1 is the primary specification. Individual controls include sex, age, and household head status. Household controls include number of adults, young children (age 0-4), and school-age children (5-17) in the household, dummies for engagement in agriculture and in enterprise, and the sex of the survey respondent. The sample is smaller in columns 5-7 because some adults are only observed once. SEs clustered at household level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Impacts of partial school reopening on adult work hours, varying sample

Sample	(1) Main	(2) Drop pre- Oct 12 R3	(3) Post = round 3	(4) Respondent	(5) Age 18-65	(6) Age 25-50	(7) Potential primary caregivers	(8) No mixed HHs	(9) Grade 6 out of ctl	(10) Grade 2 in ctl	(11) Grade 2 in, Grade 9 out of ctl	(12) Grade 12 in trt, Grades 10-11 in ctl
Post x Treat	4.32** (1.88)	4.37** (1.88)	3.27* (1.67)	5.41** (2.32)	4.89*** (1.87)	4.58** (2.13)	3.98** (2.00)	4.29** (1.88)	3.76** (1.75)	4.26** (2.01)	3.94** (2.00)	3.74* (1.99)
Post x Mixed	0.93 (1.79)	0.93 (1.79)	1.54 (1.63)	2.99 (2.14)	0.86 (1.83)	0.30 (2.01)	0.37 (1.99)		0.51 (1.96)	1.63 (1.61)	0.57 (1.62)	-0.06 (1.59)
Observations	8694	7374	8694	2997	8309	5198	5922	6700	8053	9747	9280	9929
Mean, pre-reopen control	16.21	16.65	16.44	18.88	16.27	17.26	17.46	16.21	16.38	16.21	15.98	16.21

Note: This table presents estimates of Equation 1 with varying samples. The dependent variable is total working hours over the last 7 days, taking a value of 0 for individuals not working. Observations include data from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by ‘Treat’), control households with children in an adjacent grade, and ‘Mixed’ households with both. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. SEs clustered at household level.

We first test robustness to dropping mixed households and households surveyed in round 3 before the partial reopening (which do not inform identification of treatment effects) and to changing the definition of ‘Post’ to the date the reopening was announced (between survey rounds 2 and 3). The main sample includes adults ages 18+. We then test robustness to focusing on survey respondents, adults age 18-65, adults age 25-50, and adults identified as potential parents—between 14 and 55 years older than the oldest household child—or sole caregivers (the only household adult). We also test the robustness to varying the grades included in the treatment group definitions. Omitting grade 6 students from the control definition focuses just on students in grades immediately adjacent to those treated. Including grade 2 students in the control group mirrors the inclusion of grade 6 relative to grade 8. Omitting grade 9 students prevents comparing primary to secondary school students. Finally, adding grade 12 students to the treatment definition and grade 10 and 11 students to the control definition expands the definition to include all grades eligible for the partial reopening. The definition of mixed households is updated accordingly in all cases. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Correlates of adult childcare hours in last 7 days

	(1) Childcare hours, last 7 days
Female (=1)	12.26*** (1.13)
Respondent (=1)	3.70*** (0.84)
Household head (=1)	-0.23 (0.99)
2 children in HH	0.07 (2.04)
3 children in HH	-0.18 (2.19)
4+ children in HH	1.03 (2.38)
Female $\times$ 2 children	0.37 (1.61)
Female $\times$ 3 children	2.94* (1.62)
Female $\times$ 4+ children	1.11 (1.81)
Respondent $\times$ 2 children	1.15 (1.16)
Respondent $\times$ 3 children	-0.67 (1.20)
Respondent $\times$ 4+ children	1.35 (1.26)
Household head $\times$ 2 children	-1.15 (1.42)
Household head $\times$ 3 children	1.47 (1.48)
Household head $\times$ 4+ children	0.38 (1.64)
Observations	15328
Mean, Male non-resp non-head 0 kids	10.52

Note: The table presents correlates of childcare hours in the last 7 days for all adults (age 18+) in the RRPS in survey rounds 3 and 4 when data on childcare for all adults was collected. Standard errors are clustered at the household level.

Table A10: Heterogeneity in effects on adult total work hours in the last 7 days, by sex

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Interaction term Z:	Only 1 HH child		Any child age 0-4		Any girl in grades 3-9		Any child in grades 7-9		Any child in HH ag		Above mean HH wealth	
Sample:	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women
Post x Treat, Z=0	5.94** (2.67)	3.05 (2.13)	4.20 (3.19)	3.33 (2.73)	9.64*** (2.96)	5.74** (2.75)	2.83 (3.18)	3.74 (2.75)	2.37 (2.57)	1.51 (2.47)	11.86*** (3.59)	6.01** (3.02)
Post x Mixed, Z=0	1.30 (2.44)	0.58 (1.83)	0.50 (3.71)	0.16 (2.71)	2.77 (3.67)	-0.62 (2.86)	-0.68 (5.31)	-1.63 (3.19)	-0.68 (2.55)	-0.34 (2.30)	3.91 (3.77)	4.61 (2.90)
Post x Treat, Z=1	4.02 (4.23)	4.65 (3.92)	6.34** (3.17)	3.88 (2.78)	1.59 (3.31)	1.91 (2.62)	7.65** (3.25)	3.96 (2.73)	8.99** (3.81)	6.75** (3.03)	0.97 (2.80)	1.91 (2.49)
Post x Mixed, Z=1	1.30 (2.44)	0.58 (1.83)	2.38 (2.97)	1.04 (2.34)	-0.59 (2.85)	-0.08 (2.12)	2.99 (2.85)	2.14 (2.27)	3.67 (3.57)	2.39 (2.70)	-0.85 (2.85)	-2.21 (2.20)
Observations	3887	4463	3887	4463	3887	4463	3887	4463	3887	4463	3887	4463
Mean, pre-reopen control	17.39	14.91	17.39	14.91	17.39	14.91	17.39	14.91	17.39	14.91	17.39	14.91
p-value Treated Z0=Z1	0.70	0.72	0.63	0.89	0.07	0.31	0.29	0.96	0.15	0.18	0.02	0.29
p-value Mixed Z0=Z1	.	.	0.69	0.81	0.47	0.88	0.54	0.34	0.32	0.44	0.32	0.06

Note: This table presents estimates of Equation 1 but interacting a characteristic Z with all right-hand side variables except the household fixed effects, separately for women and men. The dependent variable is total adult work hours over the last 7 days. Observations include data from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by ‘Treat’), control households with children in an adjacent grade, and ‘Mixed’ households with both. The total observations across women and men adds up to 8350 rather than 8694 as in the main analysis because some individuals only appear in their household in a subset of survey rounds. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. SEs clustered at household level.

The first columns show average effects for the full sample of households. In the following columns, coefficients with $Z = 1$ represent the sum of the $Post \times Treat$ and $Post \times Treat \times Z$ terms, and analogously for Mixed households, with standard errors calculated using the *xtlincom* package in Stata. We include p -values for tests of whether the interaction term is equal to 0. The column labels indicate which characteristic Z is being used.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Heterogeneity in effects on adult work hours outside household agriculture the last 7 days, by sex

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Interaction term Z:	Only 1 HH child		Any child age 0-4		Any girl in grades 3-9		Any child in grades 7-9		Any child in HH ag		Above mean HH wealth	
Sample:	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women
Post x Treat, Z=0	1.41 (1.63)	-1.09 (1.53)	0.64 (1.44)	0.49 (1.61)	0.42 (1.40)	-0.27 (1.63)	-0.23 (2.01)	0.30 (1.89)	0.21 (1.68)	0.08 (1.83)	2.47 (1.73)	-0.95 (1.10)
Post x Mixed, Z=0	0.31 (1.35)	-1.02 (1.07)	1.14 (1.89)	-1.77 (1.37)	1.34 (2.96)	-0.54 (2.47)	2.04 (2.47)	-1.32 (1.58)	-2.20 (1.79)	-0.55 (1.57)	1.04 (1.47)	-0.03 (1.29)
Post x Treat, Z=1	-0.99 (1.44)	3.15 (1.97)	0.48 (2.12)	-0.21 (1.81)	1.09 (1.90)	0.54 (1.68)	1.29 (1.55)	0.32 (1.50)	1.21 (1.77)	0.35 (1.42)	-0.48 (1.68)	0.64 (1.77)
Post x Mixed, Z=1	0.31 (1.35)	-1.02 (1.07)	-0.42 (1.76)	-0.07 (1.50)	-0.42 (1.53)	-0.89 (1.23)	0.23 (1.63)	-0.33 (1.30)	1.94 (1.74)	-0.51 (1.32)	-0.34 (2.05)	-1.31 (1.54)
Observations	3887	4463	3887	4463	3887	4463	3887	4463	3887	4463	3887	4463
Mean, pre-reopen control	4.57	3.44	4.57	3.44	4.57	3.44	4.57	3.44	4.57	3.44	4.57	3.44
p-value Treated Z0=Z1	0.27	0.09	0.95	0.78	0.78	0.73	0.55	0.99	0.68	0.91	0.22	0.45
p-value Mixed Z0=Z1	.	.	0.55	0.40	0.60	0.90	0.54	0.63	0.10	0.98	0.58	0.53

Note: This table replicates Table A10, but the dependent variable is hours of work outside household agriculture—in wage employment or household non-farm enterprise.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendix F: Figure Tables

Table A12: Heterogeneity by individual characteristics in effects on adult work hours in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Interaction term Z:	Female	Any work during closures	Any wage employmt during closures	Any HH ag work during closures	Any HH ent. work during closures	Ag HH	Urban location	Large urban location	Any child in private school
Post x Treat, Z=0	4.70** (2.11)	-0.77 (2.16)	3.75* (1.92)	0.26 (2.30)	4.58** (1.89)	-0.49 (3.20)	3.61 (2.75)	4.54** (2.04)	6.58*** (2.10)
Post x Mixed, Z=0	0.88 (2.09)	-1.27 (2.04)	0.95 (1.78)	-2.59 (2.39)	1.81 (1.73)	-4.25 (3.62)	0.66 (2.38)	1.27 (1.87)	0.63 (2.09)
Post x Treat, Z=1	3.99** (1.92)	4.94** (2.24)	8.79* (5.05)	4.95** (2.39)	5.63 (5.82)	5.03** (2.14)	5.30** (2.37)	1.70 (3.90)	-3.28 (3.98)
Post x Mixed, Z=1	1.02 (1.75)	1.13 (2.03)	-0.75 (5.36)	2.51 (2.12)	-9.86* (5.66)	1.66 (1.94)	1.65 (2.70)	-2.21 (4.08)	1.32 (3.40)
Observations	8694	8281	8281	8281	8281	8694	8694	8694	8694
Mean, pre-reopen control	16.21	16.13	16.13	16.13	16.13	16.21	16.21	16.21	16.21
p-value Treated Z0=Z1	0.62	0.04	0.32	0.12	0.86	0.15	0.64	0.52	0.03
p-value Mixed Z0=Z1	0.92	0.35	0.75	0.09	0.03	0.15	0.78	0.44	0.86

Note: This table presents the results shown in [Figure 4](#). The dependent variable is total working hours over the last 7 days, with individuals not working coded as working 0 hours. In each regression a characteristic Z is interacted with all right-hand side variables except the household fixed effects. Coefficients with $Z = 1$ represent the sum of the $Post \times Treat$ and $Post \times Treat \times Z$ terms, and analogously for Mixed households, with standard errors calculated using the *xtlincom* package in Stata. We include p -values for tests of whether the interaction term is equal to 0. The column label indicates which characteristic Z is being used. Columns 1-5 interact treatment with individual characteristics, and columns 6-9 interact treatment with household characteristics. Closures work participation is based on any participation in a given sector from May-October 11 2020. Samples sizes for these analyses are lower because certain adults were not listed in household rosters during any baseline round surveys, leading to missing information on closures work engagement. ‘Large urban’ is a dummy for location in one of Kenya’s largest urban areas (Nairobi, Mombasa, Nakuru, Kisumu, Kiambu) relative to any rural area, while ‘Urban’ is a dummy for location in any urban area. Private school enrollment is measured across all household children.

Observations include data from May to November 2020, and include adults age 18+ in treated households with children in grades 4 or 8 (indicated by ‘Treat’), control households with children in an adjacent grade, and ‘Mixed’ households with both. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. SEs clustered at household level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A13: Heterogeneity in effects on adult work hours in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Returning/adjacent grade child is:				Household has:	
Interaction term Z:	None	Only HH child	Sibling to age 0-4 child	A girl	In grade 7-9	Children doing HH ag work	Above mean wealth
Post x Treat	4.32** (1.88)						
Post x Mixed	0.93 (1.79)						
Post x Treat, Z=0		4.46** (2.09)	3.75 (2.64)	7.47*** (2.50)	3.34 (2.60)	1.81 (2.18)	8.82*** (3.10)
Post x Mixed, Z=0		0.87 (1.85)	0.28 (2.83)	0.69 (2.62)	-1.41 (3.65)	-0.43 (1.99)	4.31 (2.95)
Post x Treat, Z=1		4.25 (3.78)	4.93* (2.68)	1.76 (2.67)	5.66** (2.72)	7.80** (3.16)	1.37 (2.32)
Post x Mixed, Z=1		0.87 (1.85)	1.61 (2.28)	-0.35 (2.17)	2.47 (2.22)	2.93 (2.78)	-1.68 (2.15)
Observations	8694	8694	8694	8694	8694	8694	8694
Mean, pre-reopen control	16.21	16.21	16.21	16.21	16.21	16.21	16.21
p-value Treated Z0=Z1		0.96	0.76	0.12	0.54	0.12	0.05
FDR q-value Treated Z0=Z1		[0.96]	[0.87]	[0.30]	[0.89]	[0.36]	[0.22]
p-value Mixed Z0=Z1		.	0.71	0.76	0.36	0.33	0.10

Note: This table presents estimates of Equation 1 but interacting a characteristic Z with all right-hand side variables except the household fixed effects. The results are shown in Figure 5 and Figure 6. The dependent variable is total adult work hours over the last 7 days. Observations include data from May to November 2020, and include treated households with children in grades 4 or 8 (indicated by ‘Treat’), control households with children in an adjacent grade, and ‘Mixed’ households with both. ‘Post’ is a dummy for being observed on or after the partial school reopening on October 12. Regressions include household and month fixed effects. SEs clustered at household level.

The first column includes average effects for the full sample of households. In the following columns, coefficients with $Z = 1$ represent the sum of the $Post \times Treat$ and $Post \times Treat \times Z$ terms, and analogously for Mixed households, with standard errors calculated using the *xtlincom* package in Stata. We include p -values for tests of whether the interaction term is equal to 0, as well as FDR q -values for these tests calculated across outcome families for the $Post \times Treat$ interaction. The column labels indicate which characteristic Z is being used. Children engagement in HH ag work is defined based on the period when schools were fully closed. ‘Above mean wealth’ is a dummy for whether an index of household wealth, based on housing and asset ownership before the pandemic, is above the sample mean.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A14: Heterogeneity in effects on respondent childcare hours in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Returning/adjacent grade child is:				Household has:	
Interaction term Z:	None	Only HH child	Sibling to age 0-4 child	A girl	In grade 7-9	Children doing HH ag work	Above mean wealth
Post x Treat	1.51 (5.13)						
Post x Mixed	0.74 (4.77)						
Post x Treat, Z=0		-0.46 (6.13)	-3.95 (6.52)	-2.72 (6.71)	2.04 (7.77)	1.92 (5.79)	1.61 (8.59)
Post x Mixed, Z=0		1.29 (4.98)	0.83 (7.27)	3.16 (5.57)	-8.56 (7.72)	0.77 (5.47)	-3.88 (7.54)
Post x Treat, Z=1		5.34 (9.37)	8.27 (8.18)	6.27 (7.94)	0.57 (6.92)	-2.10 (10.42)	1.79 (6.32)
Post x Mixed, Z=1		1.29 (4.98)	1.26 (6.39)	-14.98 (9.92)	1.35 (6.23)	-3.78 (9.31)	3.95 (6.19)
Observations	2997	2997	2997	2997	2997	2997	2997
Mean, pre-reopen control	52.88	52.88	52.88	52.88	52.88	52.88	52.88
p-value Treated Z0=Z1		0.60	0.24	0.39	0.89	0.74	0.99
FDR q-value Treated Z0=Z1		[0.81]	[0.87]	[0.52]	[0.89]	[1.00]	[0.99]
p-value Mixed Z0=Z1		.	0.96	0.11	0.32	0.67	0.42

Note: This table replicates [Table A13](#) but the dependent variable is respondent childcare hours over the last 7 days. The results are shown in [Figure 5](#).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Heterogeneity in effects on total child hours in household agriculture in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Returning/adjacent grade child is:				Household has:	
Interaction term Z:	None	Only HH child	Sibling to age 0-4 child	A girl	In grade 7-9	Children doing HH ag work	Above mean wealth
Post x Treat	-1.03 (0.94)						
Post x Mixed	-1.01 (1.10)						
Post x Treat, Z=0		-0.34 (1.10)	-0.68 (1.24)	-2.26* (1.36)	-0.74 (1.15)	-0.48 (0.89)	0.96 (1.62)
Post x Mixed, Z=0		-0.80 (1.15)	-1.12 (1.45)	-1.76 (1.33)	-1.31 (2.14)	0.22 (1.12)	-0.23 (1.96)
Post x Treat, Z=1		-2.87* (1.68)	-1.57 (1.45)	0.44 (1.29)	-1.07 (1.48)	-3.67 (2.56)	-2.15* (1.15)
Post x Mixed, Z=1		-0.80 (1.15)	-0.91 (1.61)	0.92 (1.56)	-0.38 (1.40)	-1.47 (2.55)	-1.38 (1.26)
Observations	2997	2997	2997	2997	2997	2997	2997
Mean, pre-reopen control	3.89	3.89	3.89	3.89	3.89	3.89	3.89
p-value Treated Z0=Z1		0.21	0.64	0.15	0.86	0.24	0.12
FDR q-value Treated Z0=Z1		[0.42]	[0.87]	[0.30]	[0.89]	[0.24]	[0.24]
p-value Mixed Z0=Z1		.	0.92	0.19	0.72	0.54	0.62

Note: This table replicates [Table A13](#) but the dependent variable is total child hours in household agriculture over the last 7 days. The results are shown in [Figure 6](#).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A16: Heterogeneity in effects on hiring any agricultural labor in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Returning/adjacent grade child is:				Household has:	
Interaction term Z:	None	Only HH child	Sibling to age 0-4 child	A girl	In grade 7-9	Children doing HH ag work	Above mean wealth
Post x Treat	0.05* (0.03)						
Post x Mixed	0.02 (0.02)						
Post x Treat, Z=0		0.04 (0.03)	0.03 (0.04)	0.05 (0.04)	0.06 (0.03)	0.00 (0.03)	0.03 (0.04)
Post x Mixed, Z=0		0.02 (0.02)	0.05 (0.03)	0.02 (0.03)	-0.04 (0.05)	0.03 (0.04)	-0.02 (0.03)
Post x Treat, Z=1		0.07 (0.06)	0.07* (0.04)	0.05 (0.04)	0.04 (0.04)	0.11** (0.05)	0.08** (0.04)
Post x Mixed, Z=1		0.02 (0.02)	-0.00 (0.03)	0.06 (0.05)	0.04 (0.03)	0.02 (0.03)	0.05 (0.03)
Observations	2997	2997	2997	2997	2997	2997	2997
Mean, pre-reopen control	0.07	0.07	0.07	0.07	0.07	0.07	0.07
p-value Treated Z0=Z1		0.71	0.49	0.92	0.82	0.07	0.36
FDR q-value Treated Z0=Z1		[0.81]	[0.49]	[0.93]	[0.89]	[0.04]	[0.32]
p-value Mixed Z0=Z1		.	0.30	0.41	0.13	0.86	0.11

Note: This table replicates [Table A13](#) but the dependent variable is a dummy variable for hiring any agricultural workers over the last 7 days. The results are shown in [Figure 6](#).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A17: Heterogeneity in effects on adults hours outside household agriculture in the last 7 days

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Returning/adjacent grade child is:				Household has:	
Interaction term Z:	None	Only HH child	Sibling to age 0-4 child	A girl	In grade 7-9	Children doing HH ag work	Above mean wealth
Post x Treat	0.49 (0.95)						
Post x Mixed	-0.29 (0.86)						
Post x Treat, Z=0		0.17 (1.21)	0.63 (1.21)	0.01 (1.08)	0.16 (1.62)	0.18 (1.40)	0.70 (1.05)
Post x Mixed, Z=0		-0.42 (0.90)	-0.31 (1.19)	0.22 (1.83)	0.16 (1.51)	-1.24 (1.28)	0.50 (0.94)
Post x Treat, Z=1		1.27 (1.54)	0.16 (1.54)	0.84 (1.46)	0.80 (1.10)	0.84 (1.29)	0.18 (1.40)
Post x Mixed, Z=1		-0.42 (0.90)	-0.25 (1.24)	-0.68 (1.05)	-0.12 (1.11)	0.70 (1.12)	-0.89 (1.35)
Observations	8694	8694	8694	8694	8694	8694	8694
Mean, pre-reopen control	4.08	4.08	4.08	4.08	4.08	4.08	4.08
p-value Treated Z0=Z1		0.58	0.81	0.64	0.74	0.73	0.77
p-value Mixed Z0=Z1		.	0.97	0.67	0.88	0.25	0.40

Note: This table replicates [Table A13](#) but the dependent variable is adult hours of work outside household agriculture—in household non-farm enterprise or wage employment—over the last 7 days, with individuals not working coded as working 0 hours.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$